DNA sequencing

Christiaan Henkel
Bastienne Vriesendorp



Universiteit
Leiden

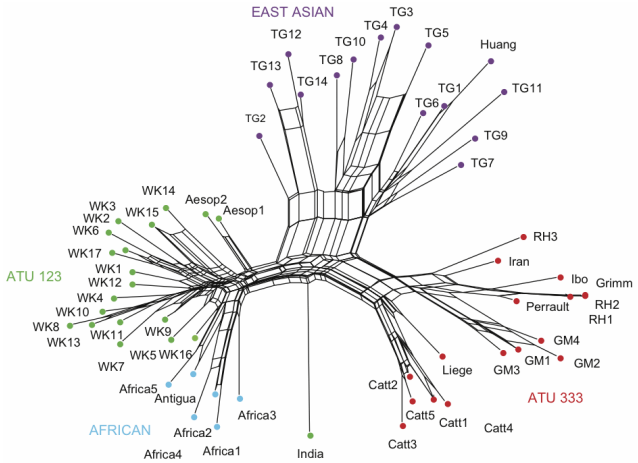
MSc course Genomic Architecture 3-11-2025

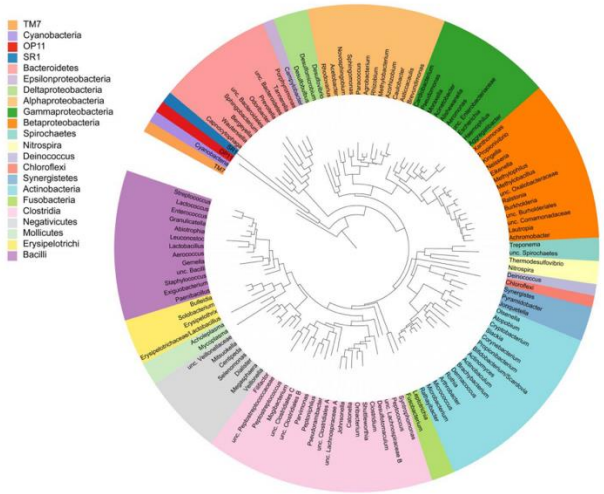
Background

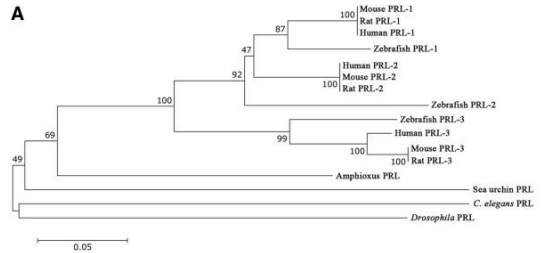


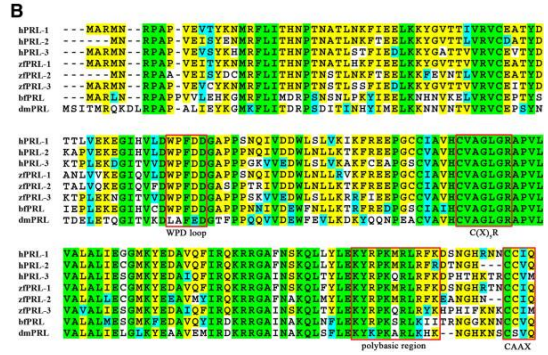
Christiaan Henkel
Genome scientist at Norwegian University of Life Sciences in Ås, near Oslo.
At IBL: courses on Bioinformatics and genomics

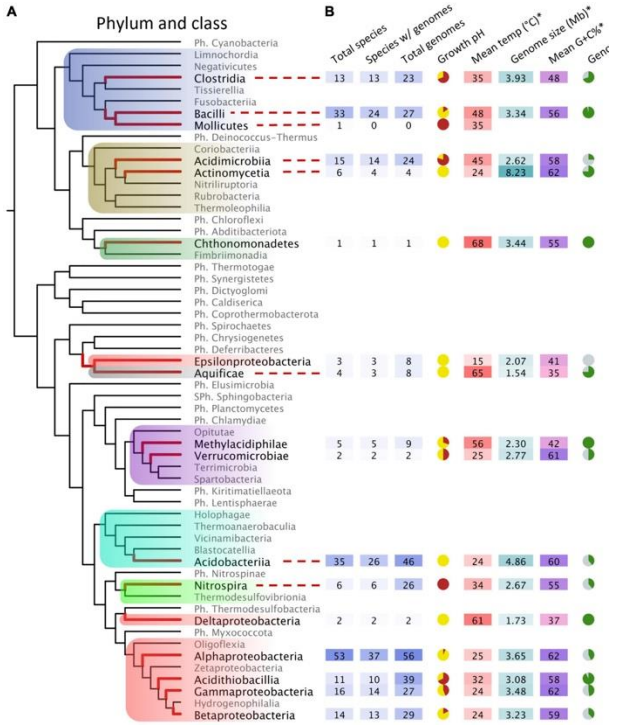
Bastienne Vriesendorp
Since 2021 at IBL: Facility manager bioinformatics, teacher, coordinator Bioinformatics BSc



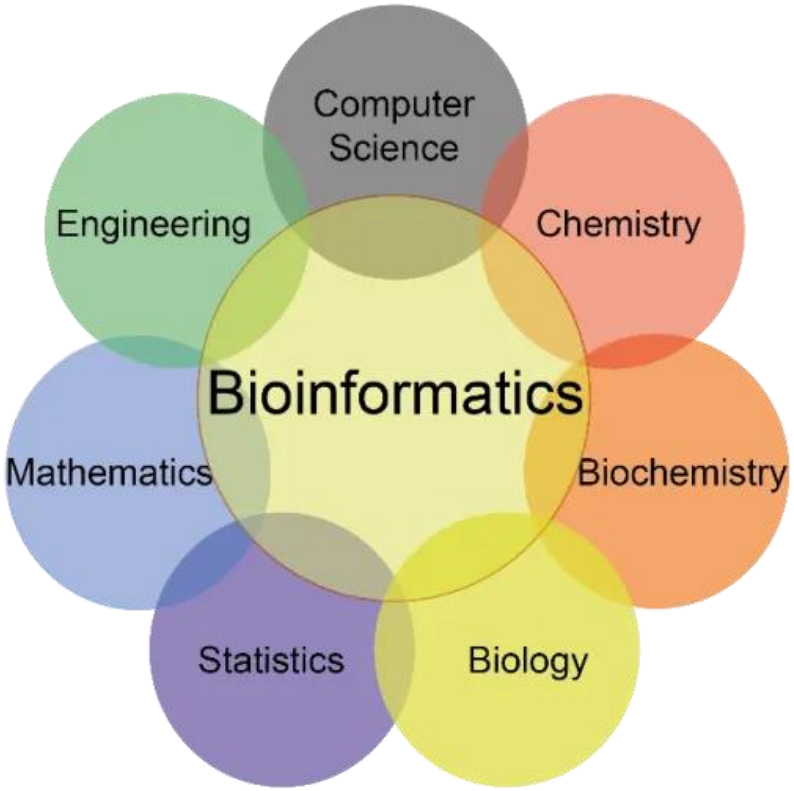




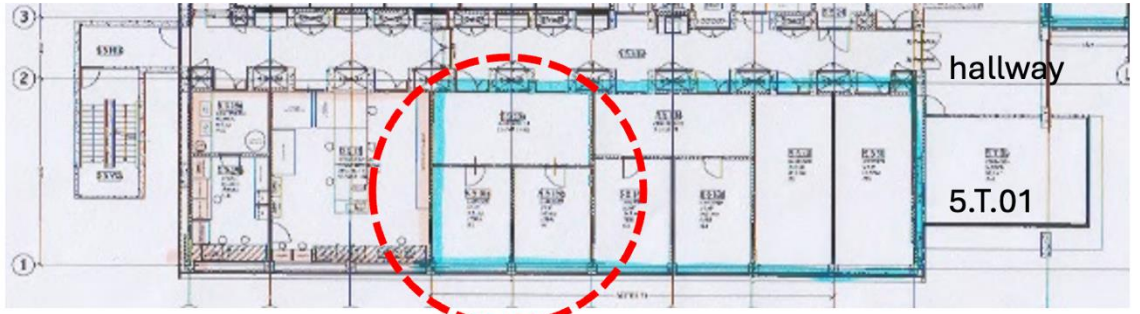




Bioinformatics at IBL



Bioinformatics Atelier at 5th floor



5.5.16/5.5.15

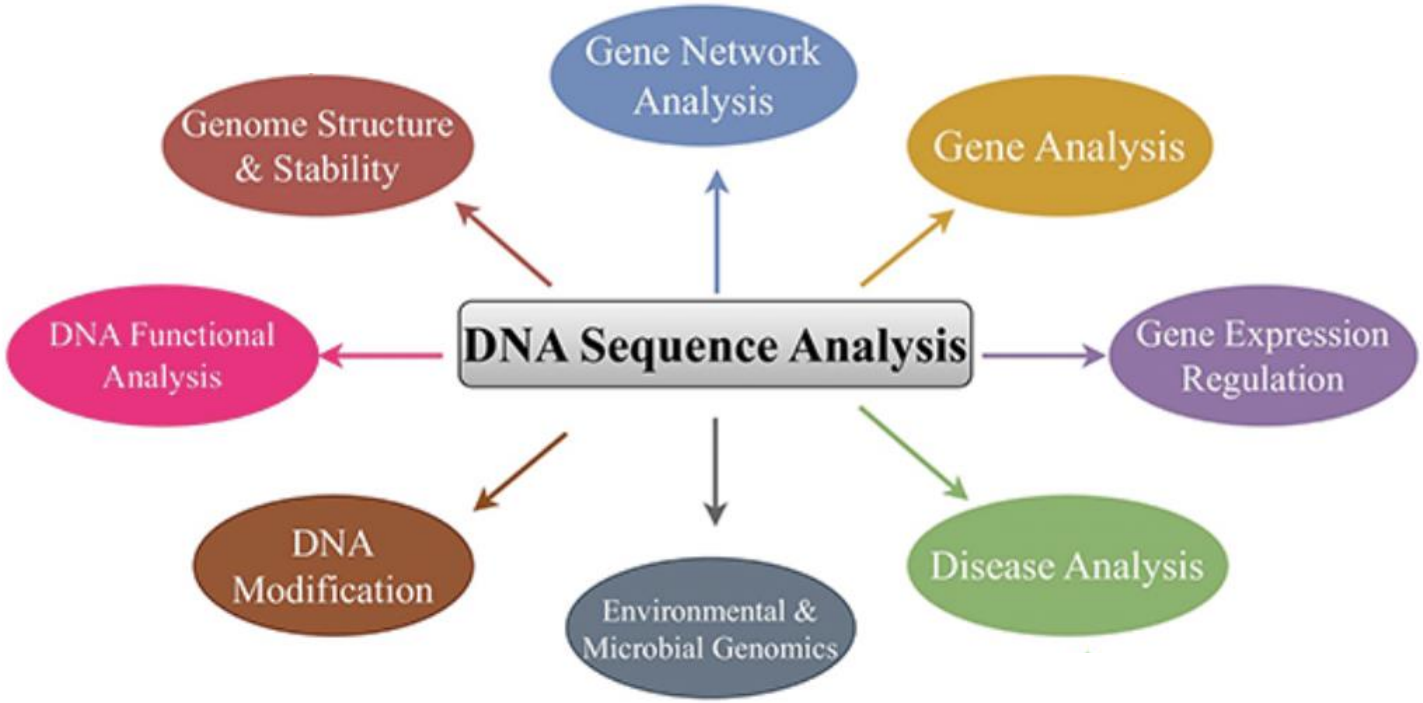


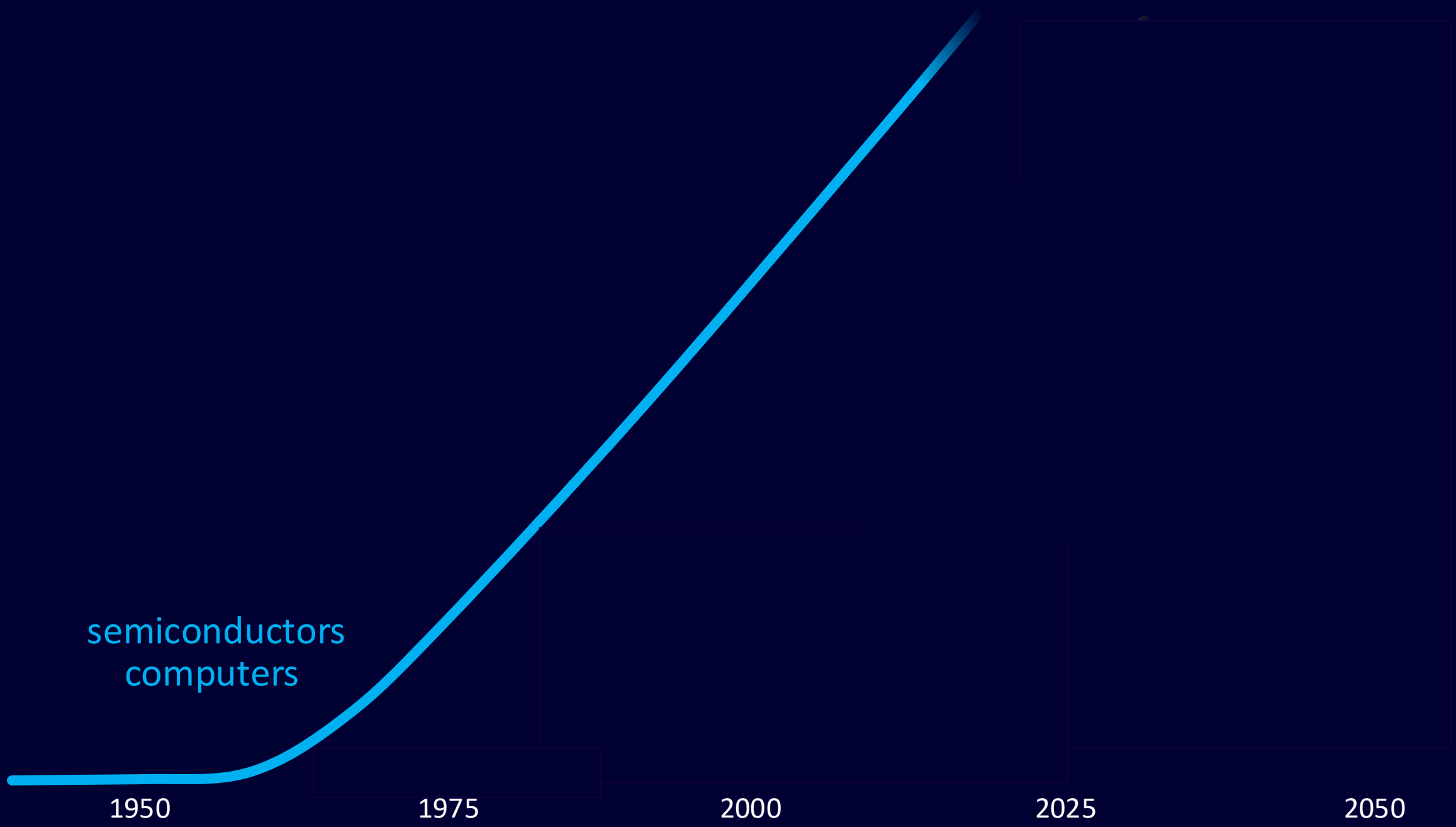
Chao Du



Belmin Bajramovic

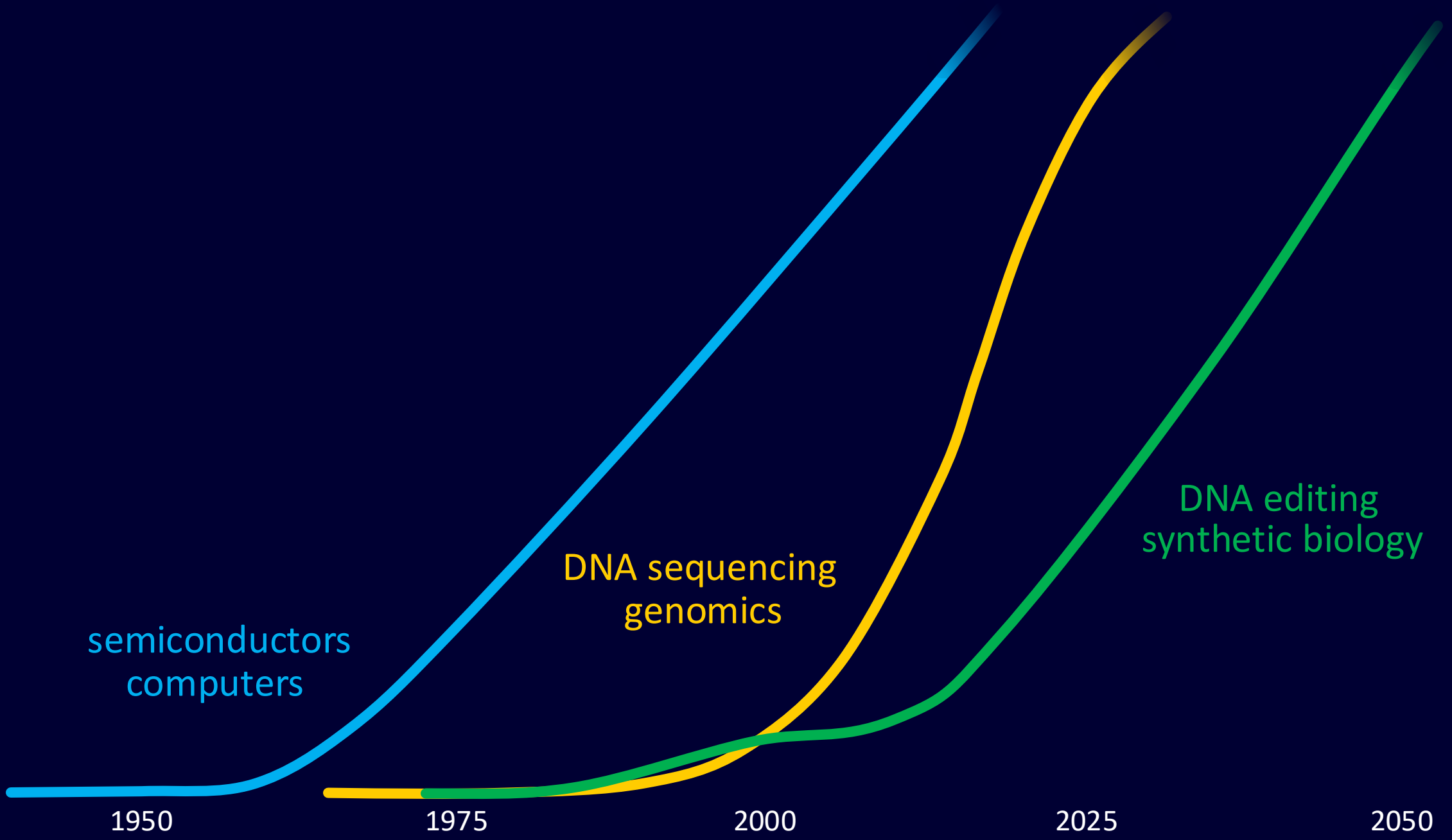
Importance of DNA sequencing in biology

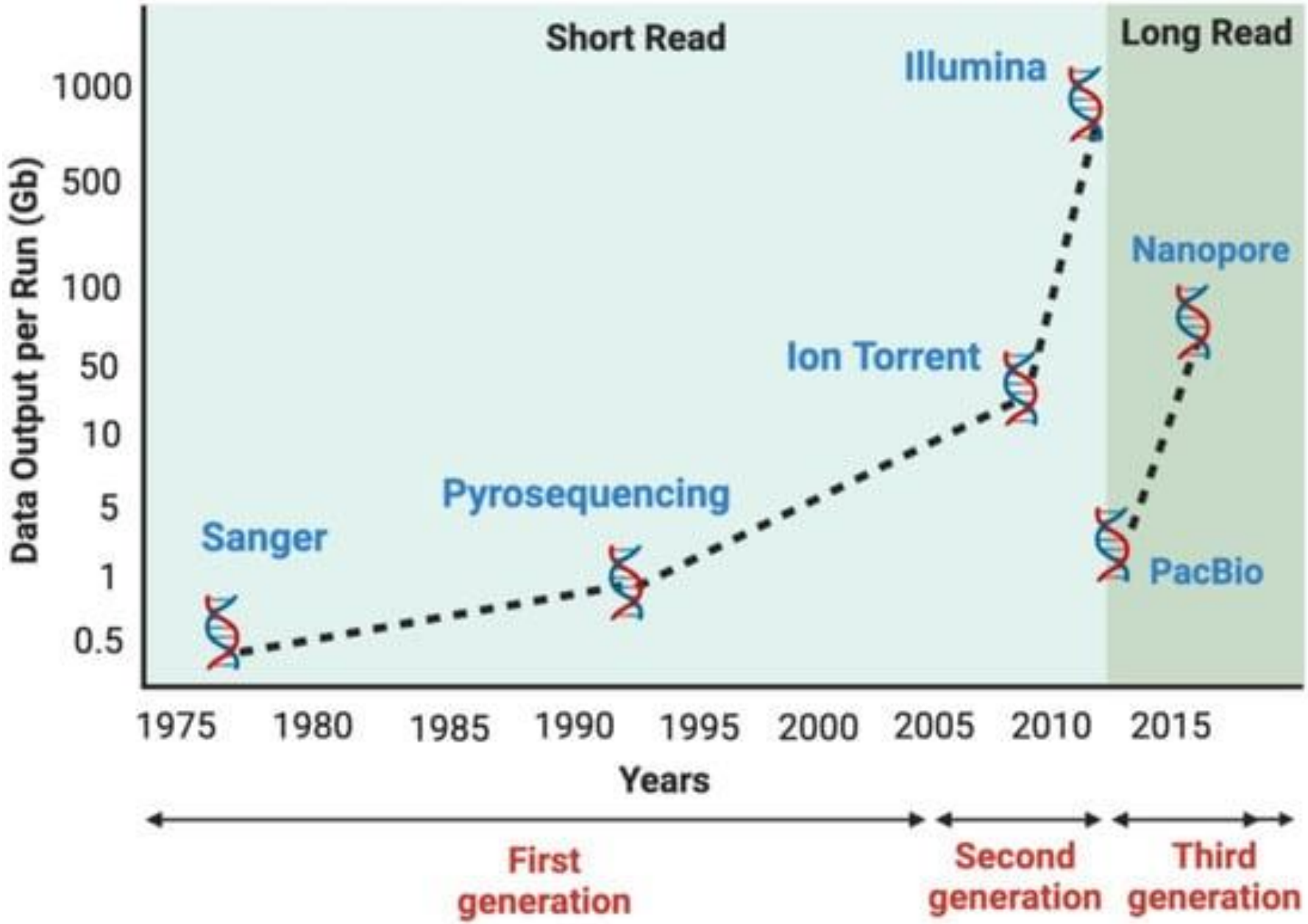


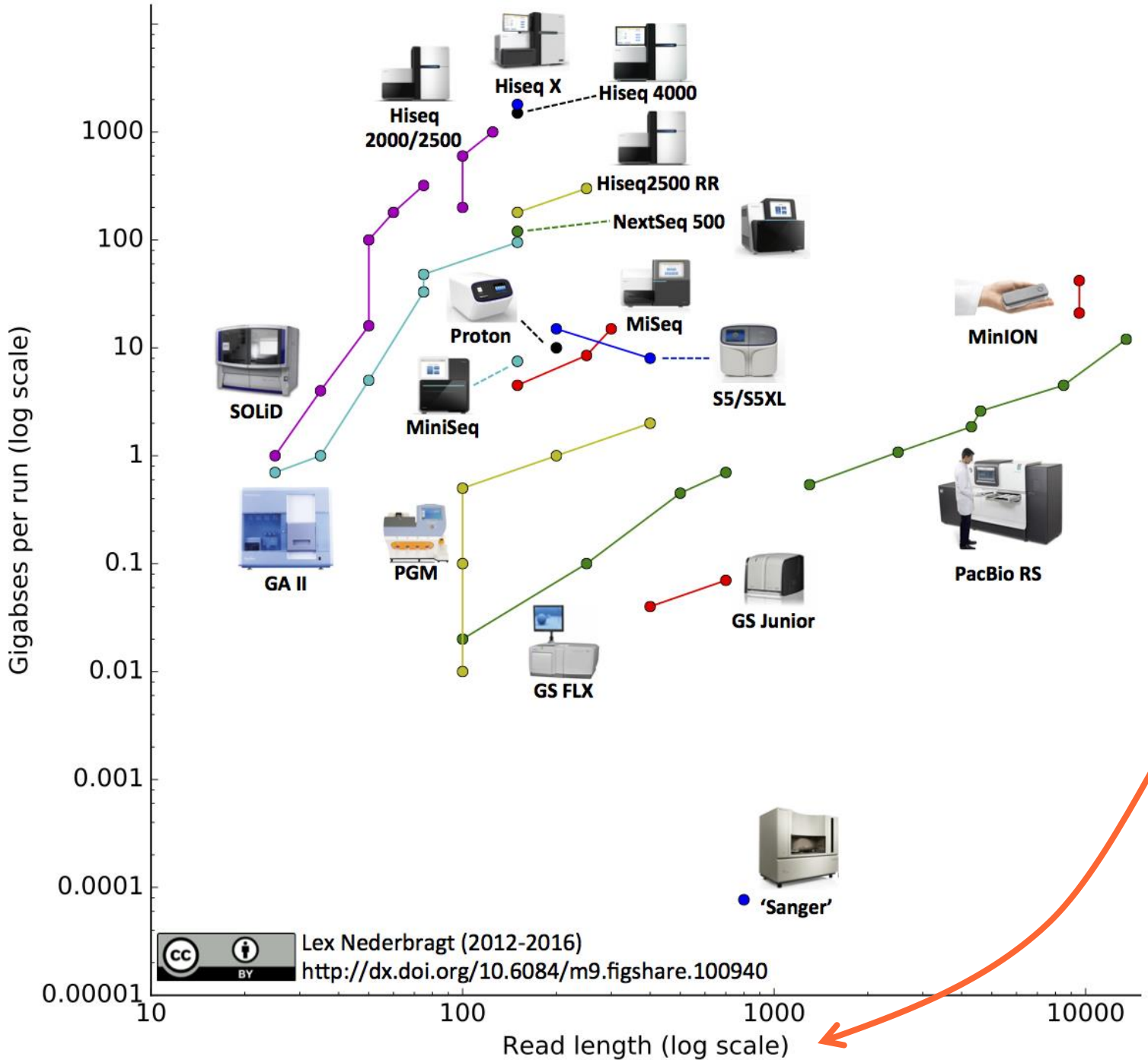

semiconductors
computers
1950
1975
2000
2025
2050





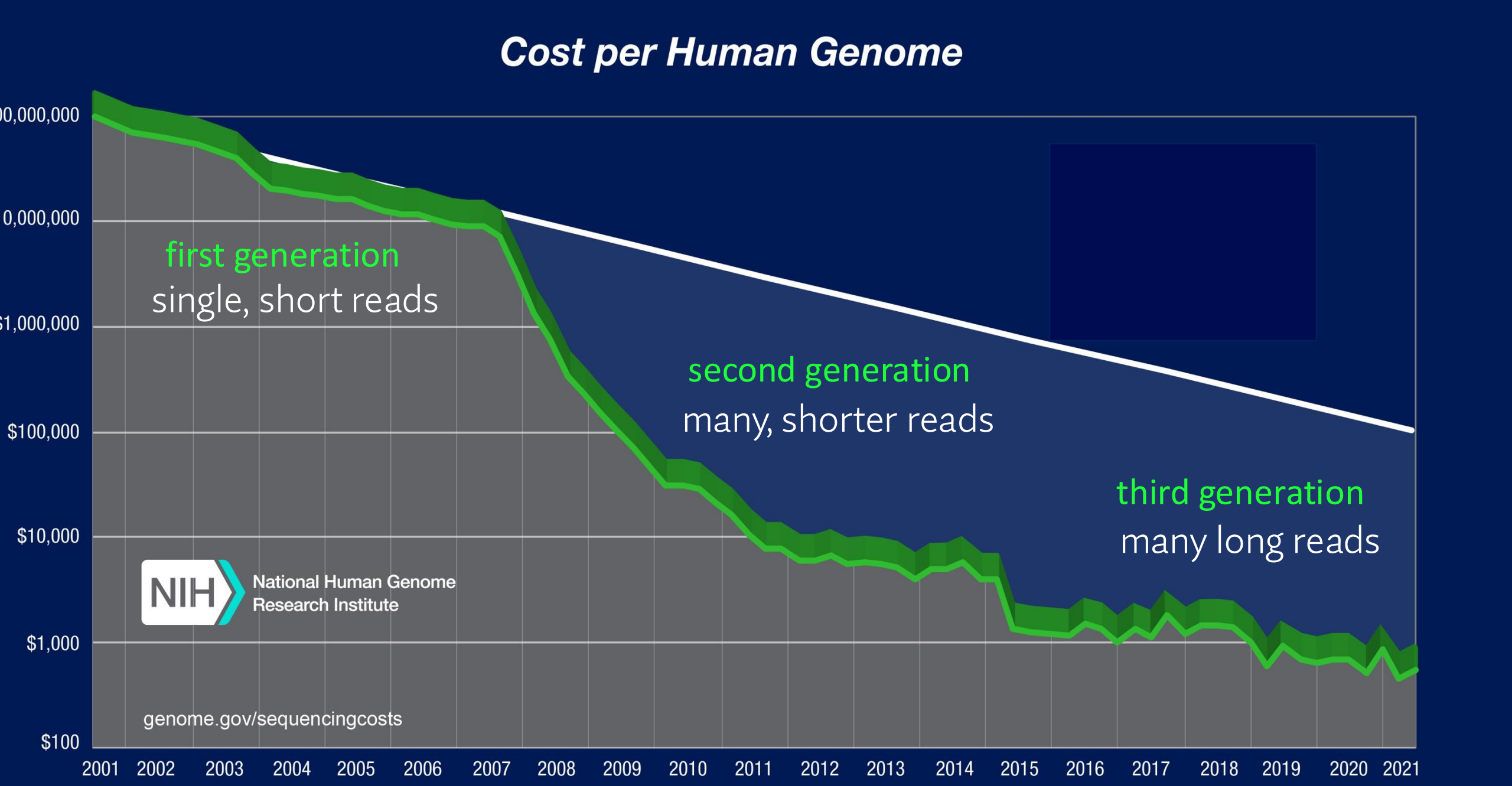

semiconductors
computers
DNA sequencing
genomics
DNA editing
synthetic biology
1950
1975
2000
2025
2050


Short Read
Long Read
Illumina
Nanopore
Ion Torrent
Pyrosequencing
Sanger
PacBio
Data Output per Run (Gb)
1000
500
100
50
10
5
1
0.5
1975
1980
1985
1990
1995
2000
2005
2010
2015
Years
First generation
Second generation
Third generation



sequencing read:
the result of sequencing a single DNA fragment

(a computer text file of ATCGs)


Cost per Human Genome
0,000,000
0,000,000
$1,000,000
$100,000
$10,000
$1,000
$100
first generation
single, short reads
second generation
many, shorter reads
third generation
many long reads
NIH
National Human Genome
Research Institute
genome.gov/sequencingcosts
2001
2002
2003
2004
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
2017
2018
2019
2020
2021

DNA sequencing generations

First generation

- Sanger sequencing (1977)

Second generation (next generation sequencing, NGS)

- 454 pyrosequencing (2005)/Thermo Fisher Ion Torrent (2010)
- Solexa/Illumina sequencing by synthesis (2005)
- Complete Genomics/MGI DNA nanoballs (2006?)
- Life Technologies SOLiD (2006)
- Element avidity (2023)

Third generation (also NGS)

- Pacific Biosciences single-molecule, real-time sequencing (2011)
- Oxford Nanopore Technologies sequencing (2014)

Sanger sequencing

Principle: use DNA polymerase to copy the DNA, incorpate chain-terminating agents to stop DNA synthesis, generating many DNA strands of varying lenghts.

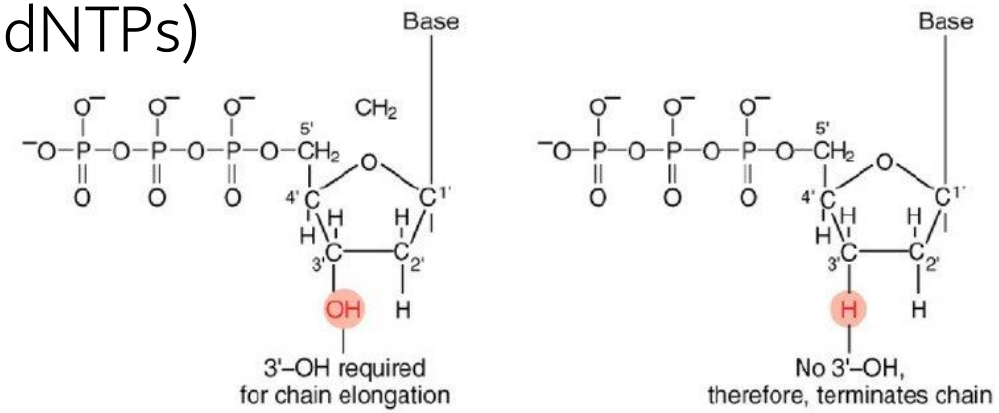




Frederick Sanger

Sanger sequencing

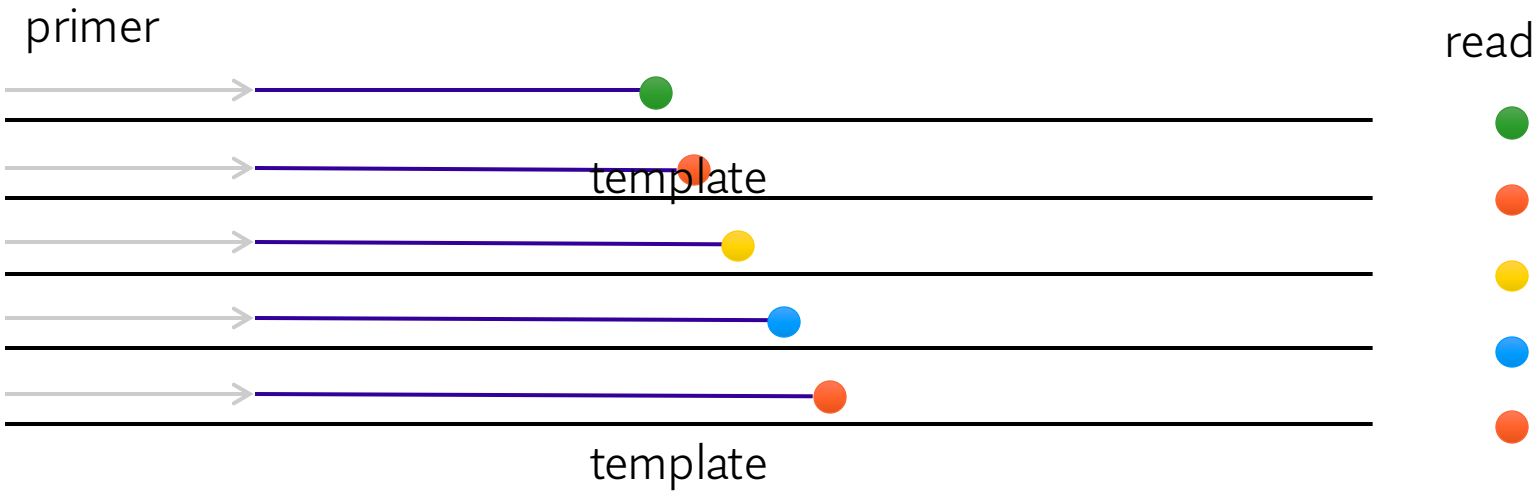
Based on chain termination by dideoxynucleotides (ddNTPs)



Example: DNA amplification in the presence of dNTP and ddATP

Template	3’ ATCGCGATGCTAGCATGAACTGTCGA
Copy	5’ TAGCGCTACGATCGTACTTGACAGCT
Fragments	5’ TAGCGCTACGATCGTACTTGACA
	5’ TAGCGCTACGATCGTACTTGA
	5’ TAGCGCTACGATCGTA
	5’ TAGCGCTACGA

Sanger sequencing

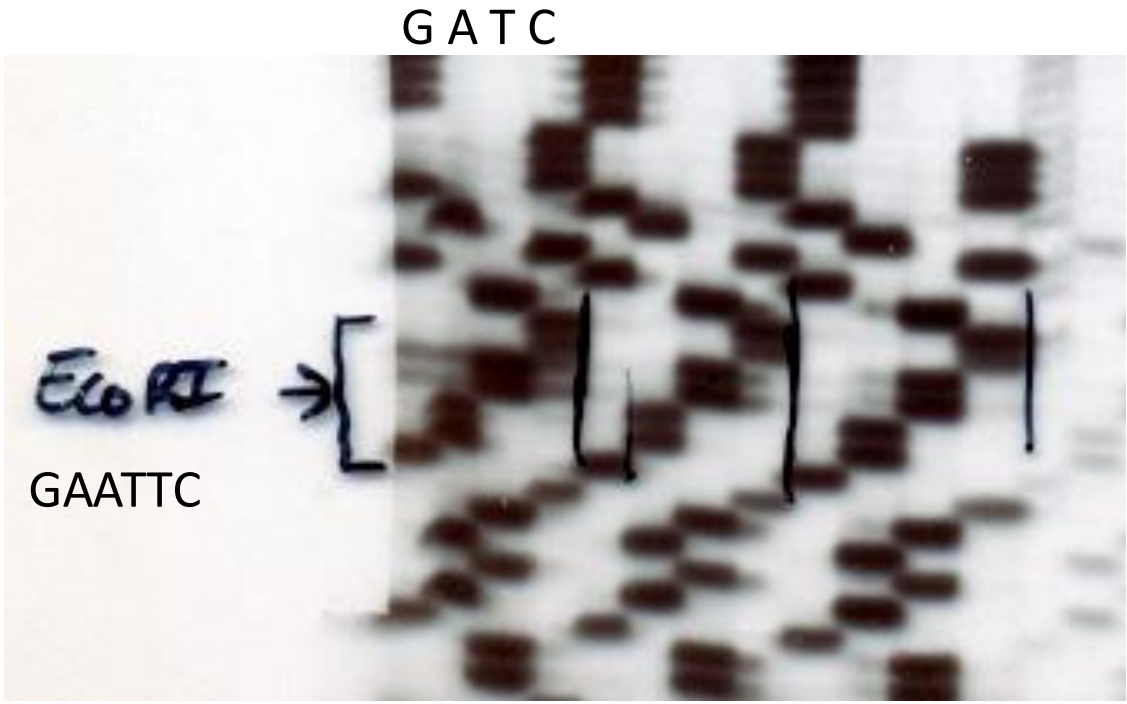


Experimental challenge:
disentangle the signal consisting of four nucleotides and many fragment lengths

Sanger's solution

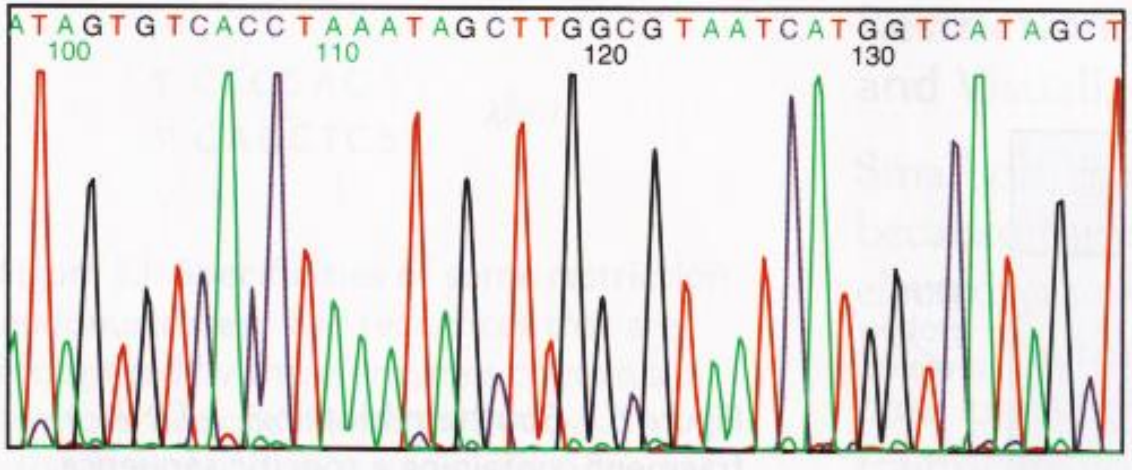
Sanger's solution:

- Four separate reaction volumes (ddATP, ddTTP, ddCTP, ddGTP)
- Radioactive labels
- Length separation by gel electrophoresis
- Sequencing read: four lanes



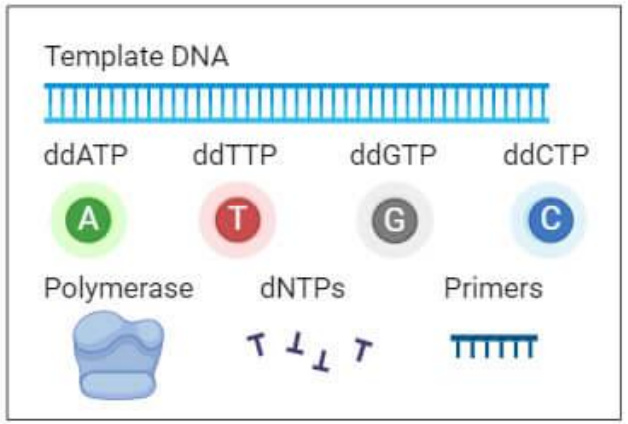
Automatic:

- A single reaction volume
- Fluorescent labels
- Length separation by gel electrophoresis

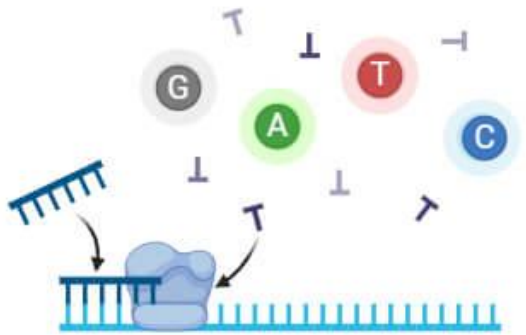


Sanger Sequencing

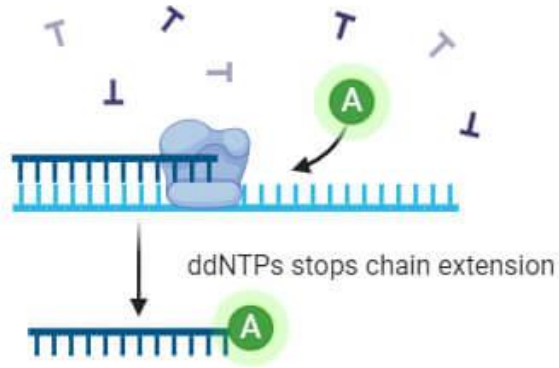
Reagents



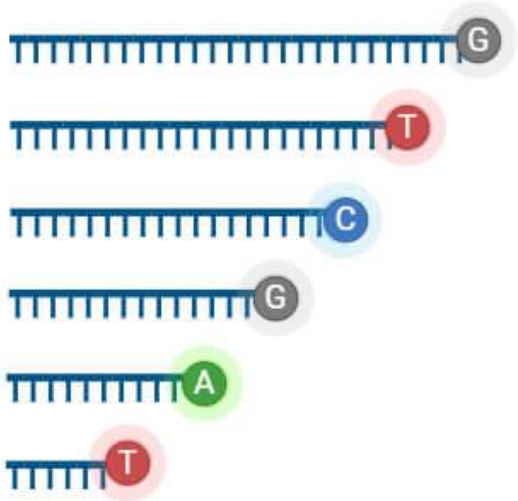
① Primer annealing and chain extension



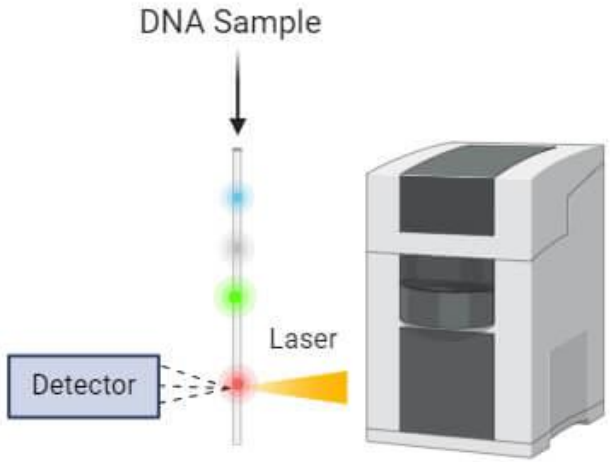
② ddNTP binding and chain termination



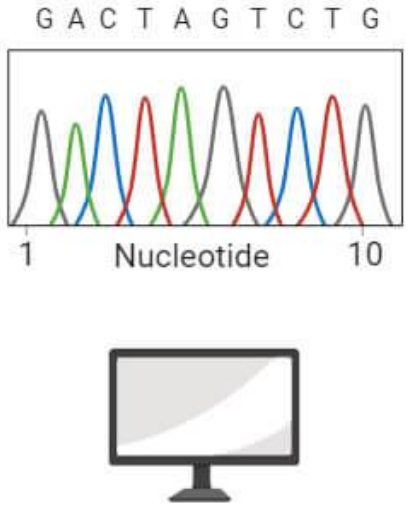
③ Fluorescently labelled DNA sample



④ Capillary gel electrophoresis and fluorescence detection



⑤ Sequence analysis and reconstruction



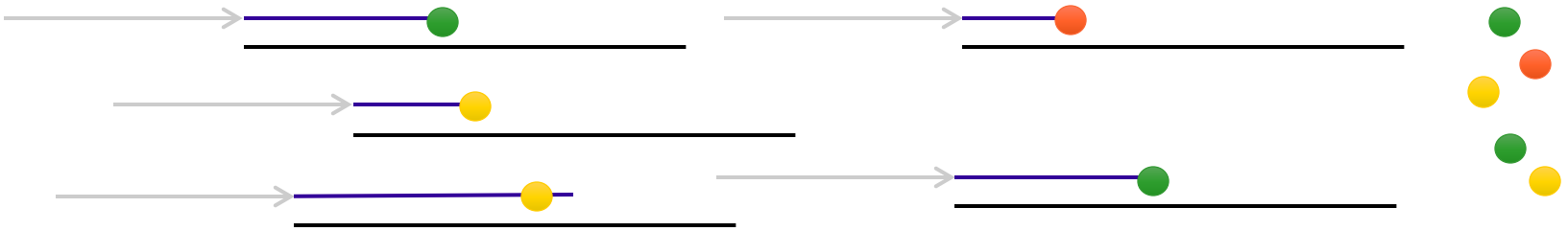
Applied Biosystems *3730 xl*

- Samples in 96, 384 wells plates (stack of 16)
- 96 capillaries (36–50 cm)
- Read length 400–900 bp
- Throughput max. 2.88 Mbp/day



2nd generation (NGS) sequencing

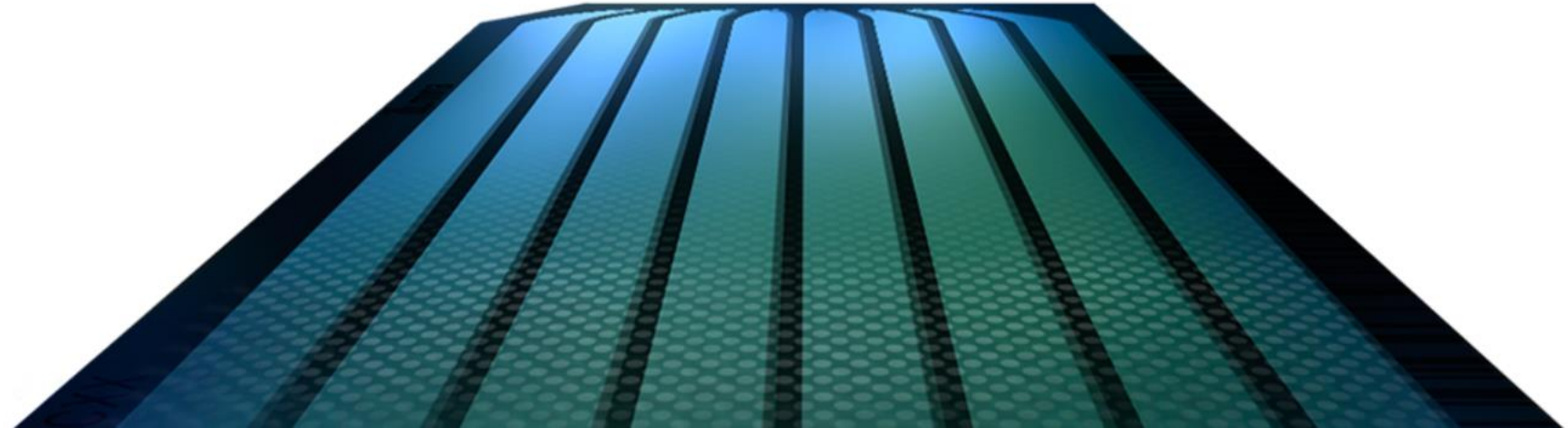
- A sequencing library consists of many, small, ‘randomly’ generated DNA fragments
- A universal primer template (an **adapter sequence**) must be attached to every fragment
- Many (millions to billions) simultaneous, independent sequencing reactions
- Signal confusion!

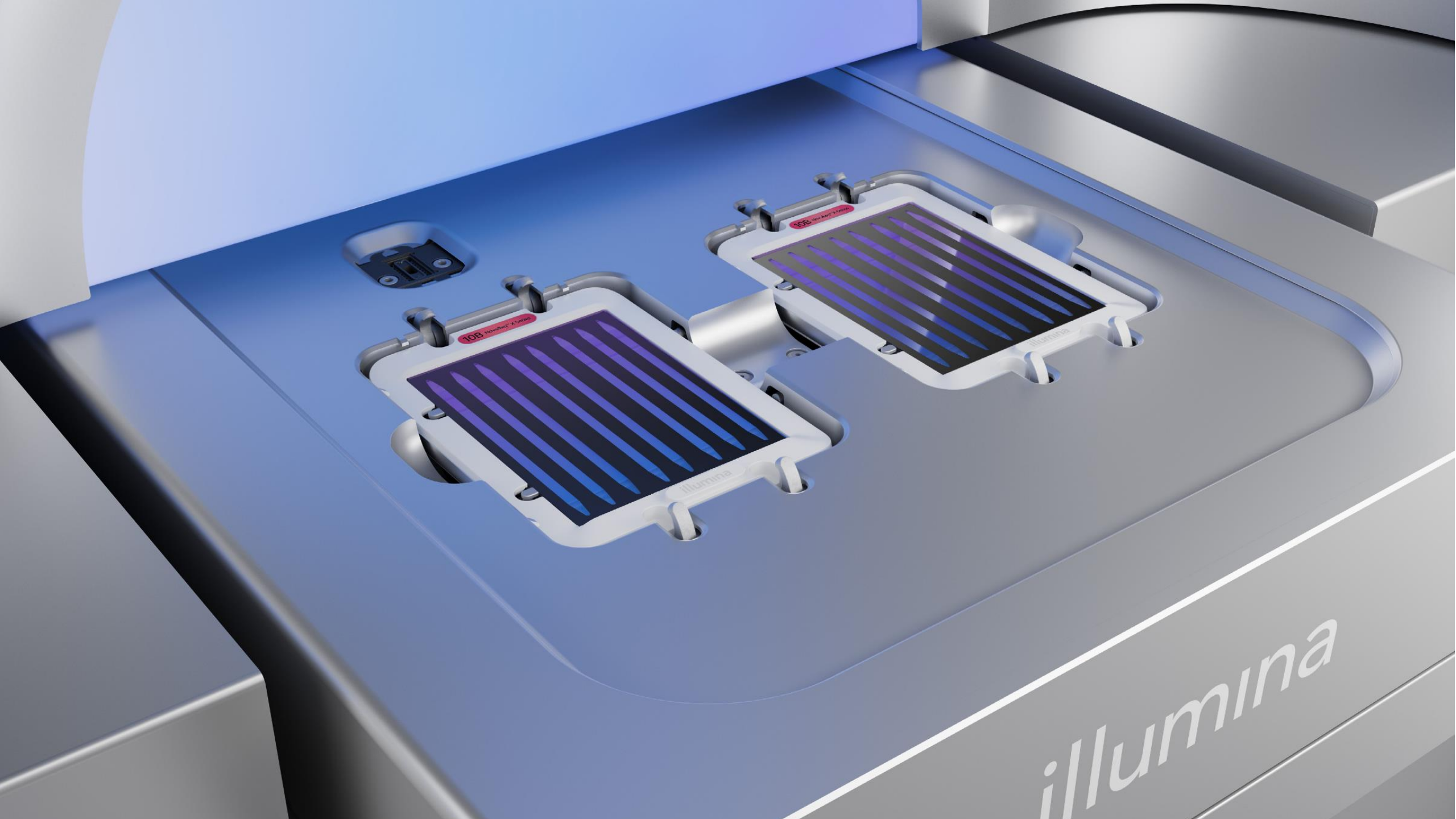


illumina®

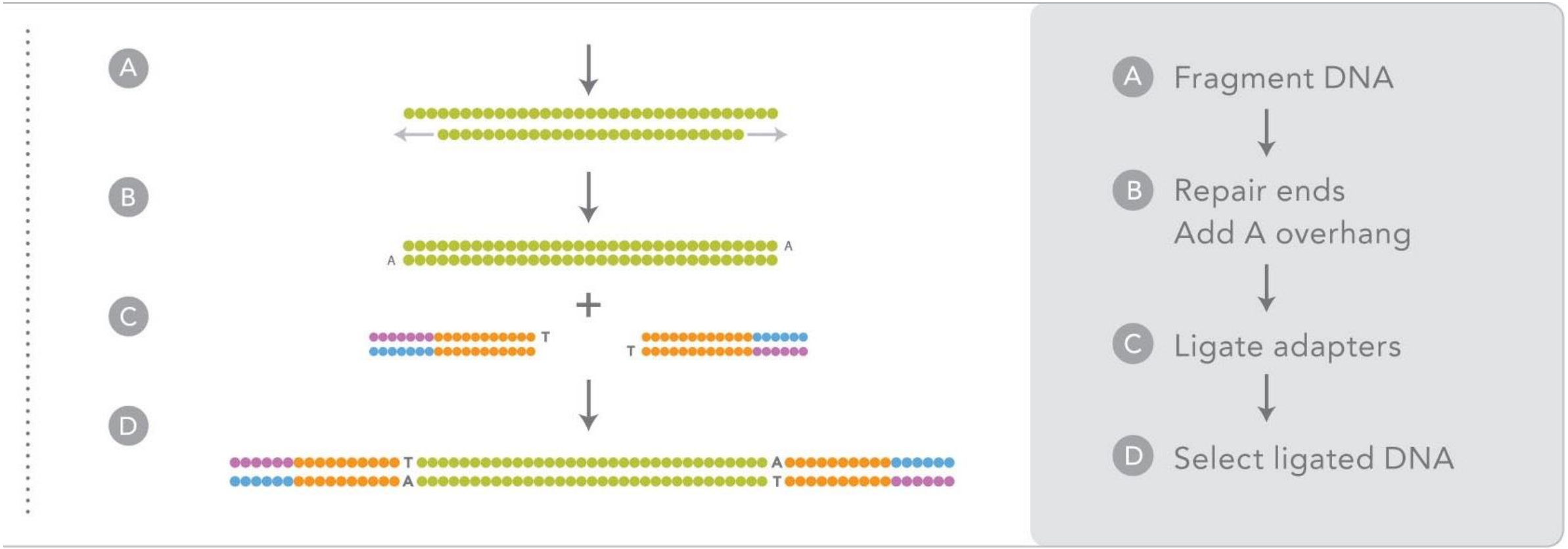


Illumina: “sequencing by synthesis (SBS)”

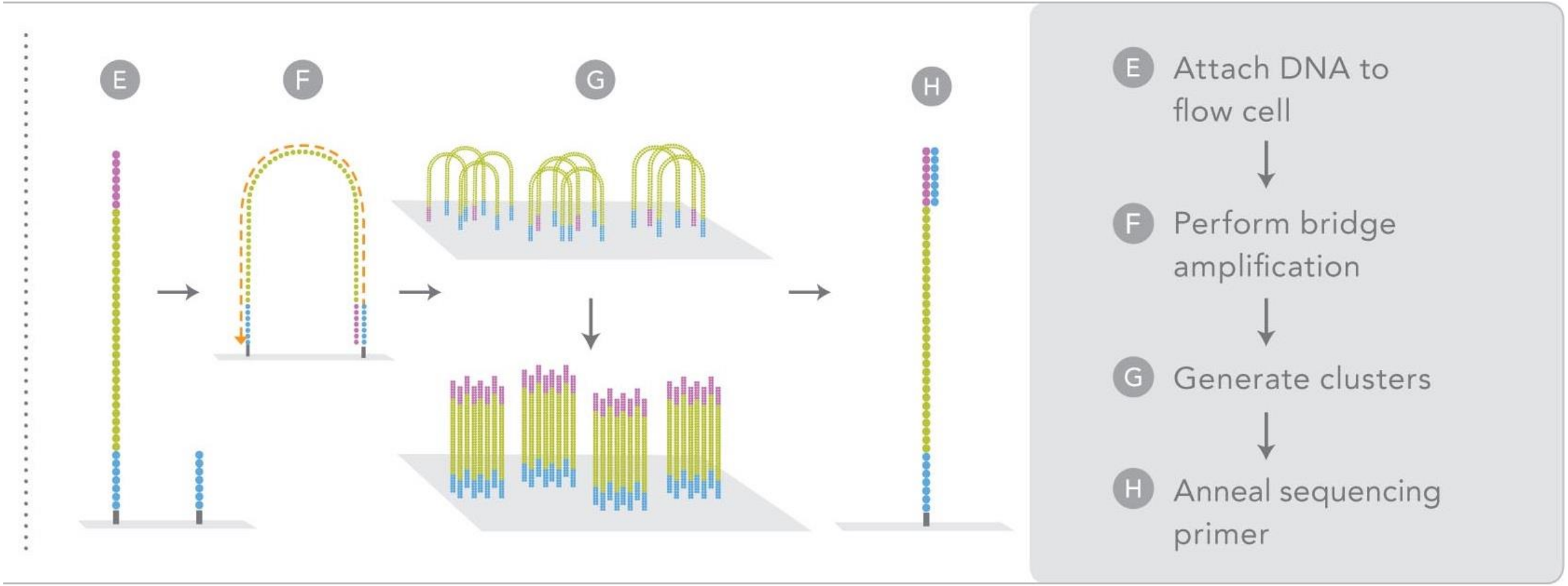



illumina

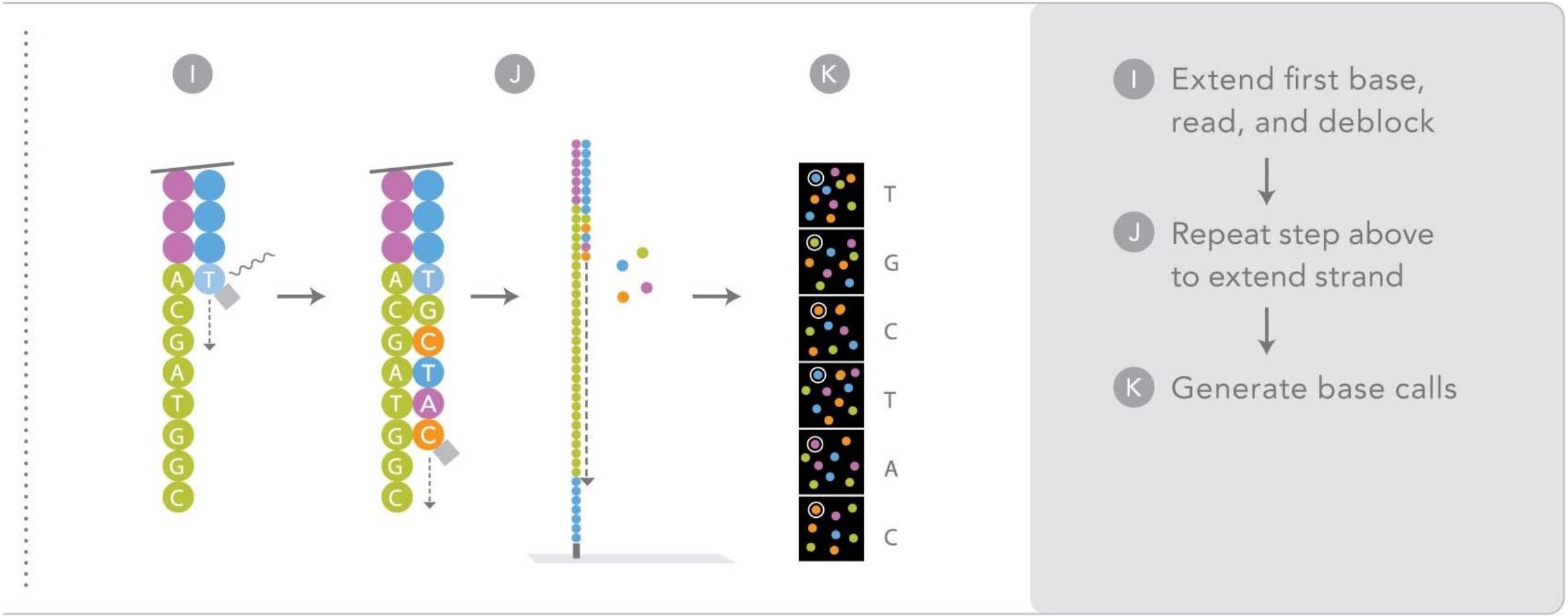
Illumina sequencing: library prep



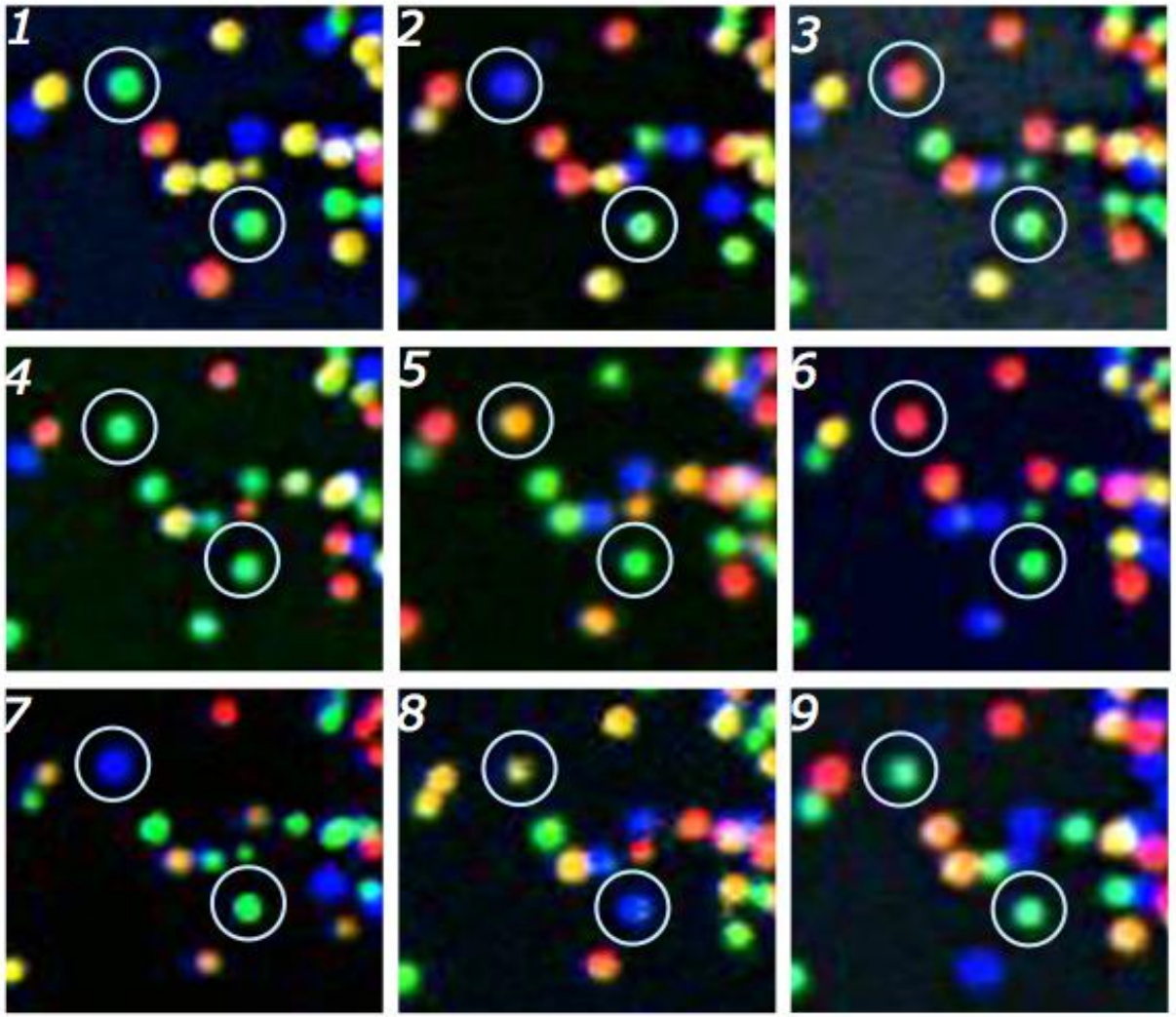
Illumina sequencing: amplification



Illumina SBS sequencing & signal detection

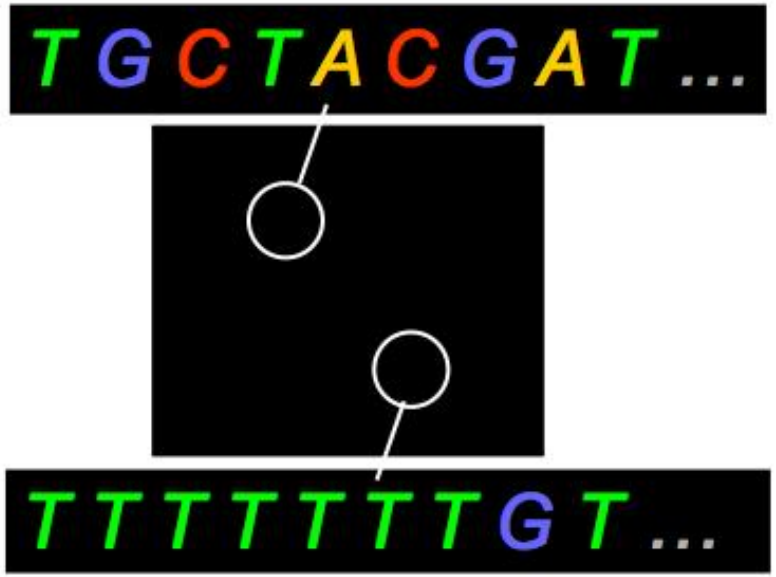


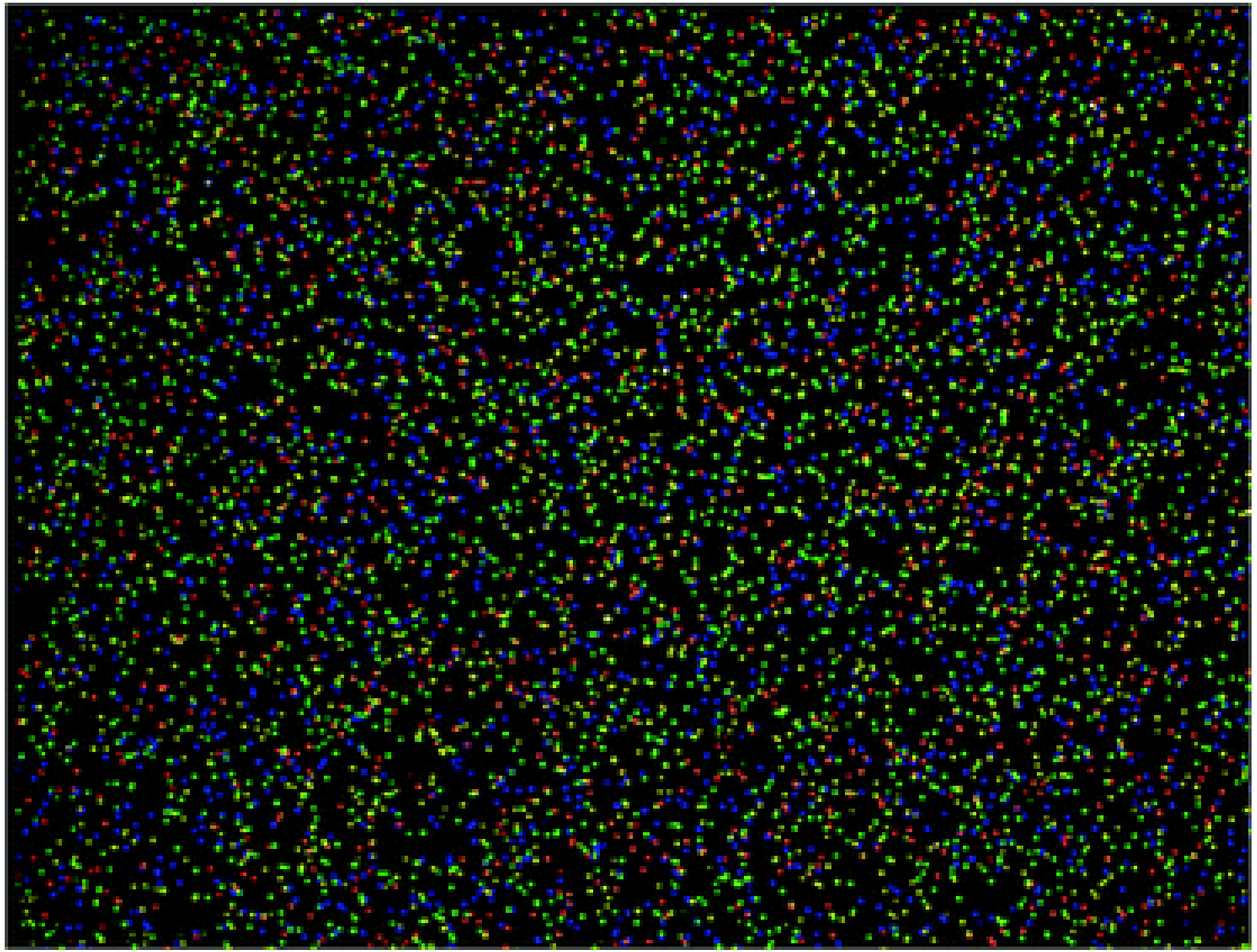
Illumina sequencing: signal detection



Solexa

illumina





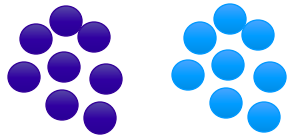
DOIhttps://doi.org/10.1039/C1CC11078K

Illumina sequencing instruments

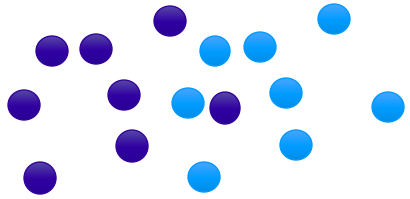
	 iSeq 100 System	 MiniSeq System	 MiSeq Series ⊕	 NextSeq Series ⊕	 HiSeq 4000 System	 HiSeq X Series‡	 NovaSeq 6000 System
Run Time	9–17.5 hrs	4–24 hours	4–55 hours	12–30 hours	< 1–3.5 days	< 3 days	~13 - 38 hours (dual SP flow cells) ~13–25 hours (dual S1 flow cells) ~16–36 hours (dual S2 flow cells) ~44 hours (dual S4 flow cells)
Maximum Output	1.2 Gb	7.5 Gb	15 Gb	120 Gb	1500 Gb	1800 Gb	6000 Gb
Maximum Reads Per Run	4 million	25 million	25 million †	400 million	5 billion	6 billion	20 billion
Maximum Read Length	2 × 150 bp	2 × 150 bp	2 × 300 bp	2 × 150 bp	2 × 150 bp	2 × 150 bp	2 x 250**

Illumina reads are intrinsically short

- Amplification is less reliable for longer fragments
- Trade-off between fragment length and spot density (nr. of reads)



short fragments



very long fragments

- Sequencing reactions are never 100% efficient
- Therefore, after many sequencing ‘cycles’, fragments in spots go out of sync



cycle 1



cycle 50



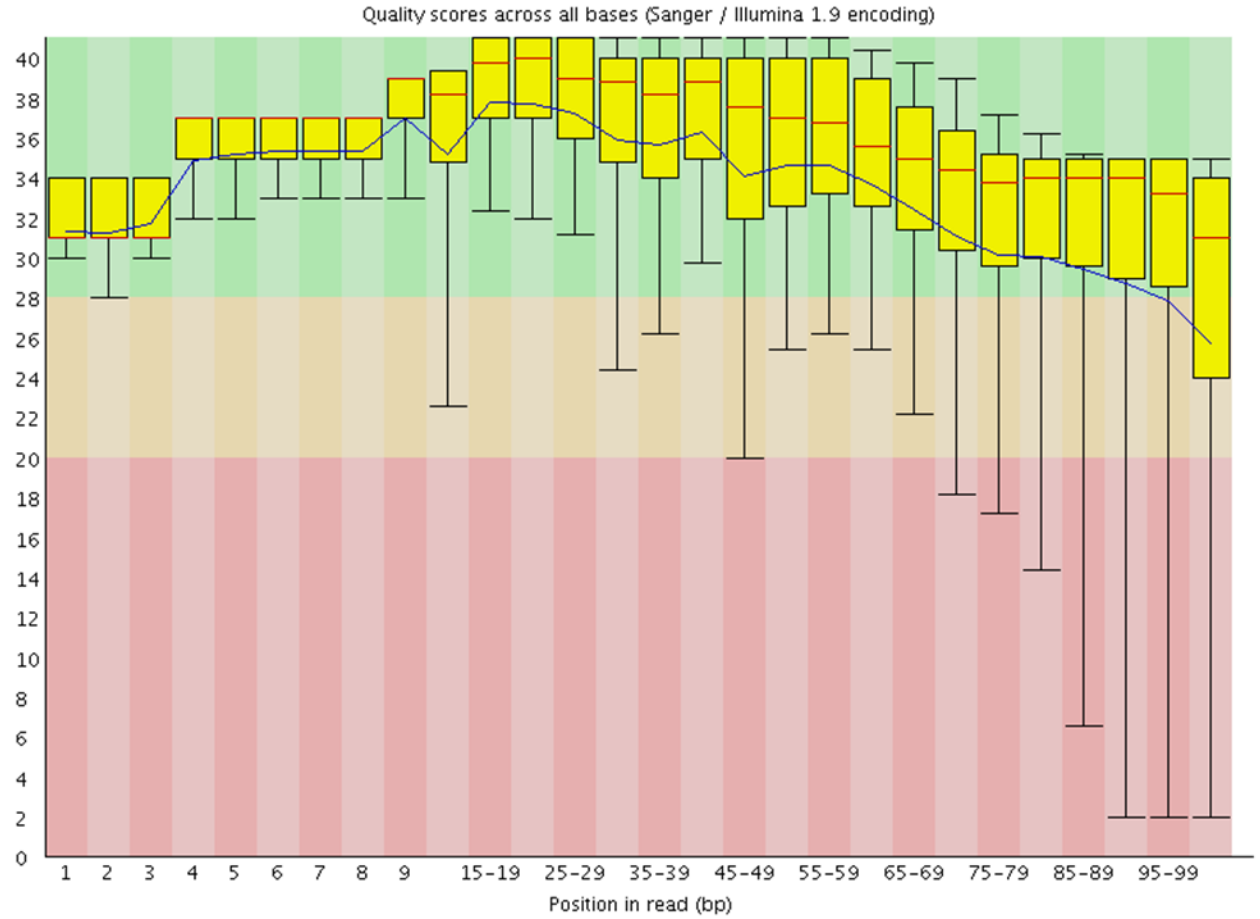
cycle 100



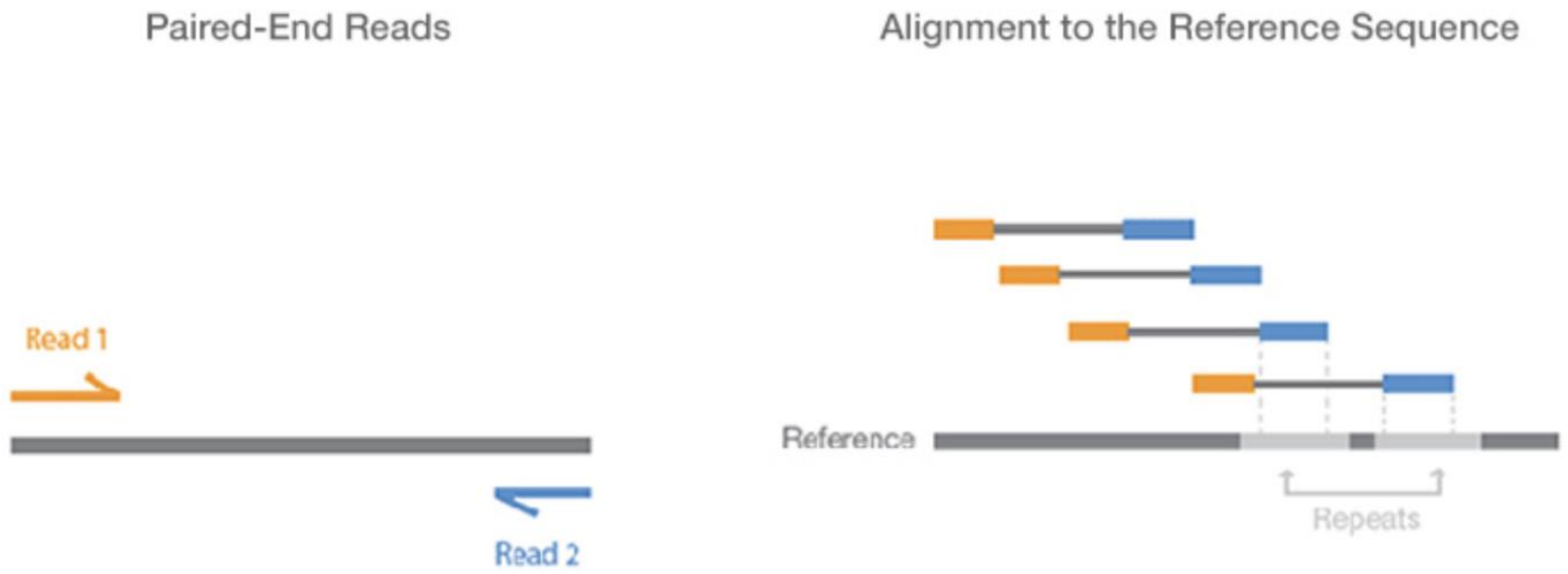
cycle 250

Illumina reads are intrinsically short

- Illumina read quality: typically >99.9% of nt identified correctly (Q30)
- Quality decreases along the read



Illumina sequencing: paired ends



Illumina sequencing summary

😃 Extremely high throughput (16 Tbp per NovaSeq X run: 16000000000000 bp)

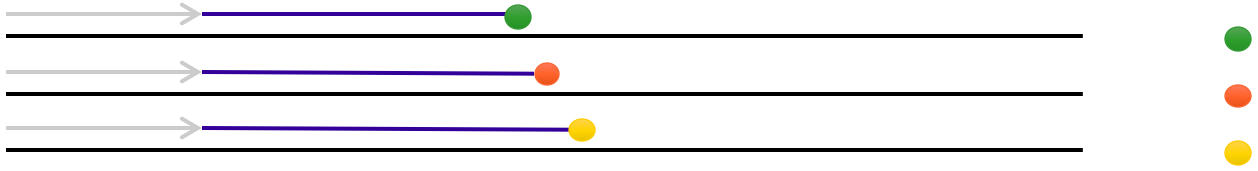
🙂 High quality data (>99.9% sequence identity, **Q30**)

😕 Very short reads (usually up to 2×150 bp)

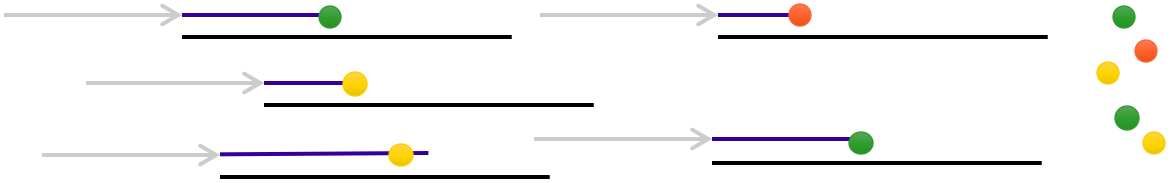
- Reads are much shorter than in first-generation sequencing (<1 kbp)
- Short reads are great (😃) for **quantitative applications**
  (e.g. counting the numbers of a specific DNA/RNA species)
- Longer reads are desirable for **structural applications**
  (e.g. genome sequencing, transcript isoforms)

Sanger vs. Illumina

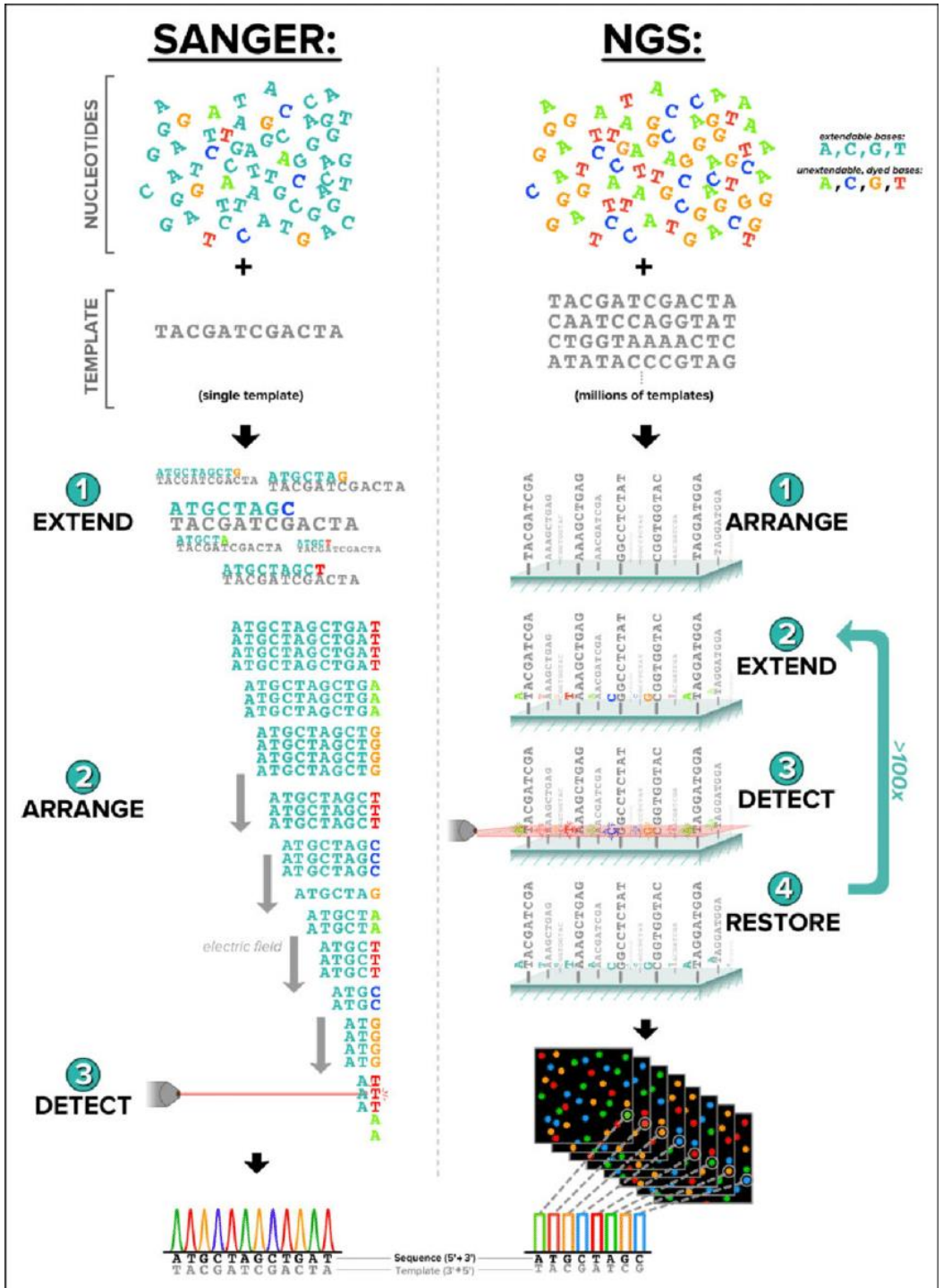
- Both Sanger and second-generation sequencing require amplification of DNA prior to sequencing (signal boost)
- Sanger: single long fragment



- Next-generation sequencing: many short fragments (a sequencing library)



- Sanger: accurate and reliable, good for validation
- Illumin: fast and cost-effective



Third-generation sequencing: no amplification

- Amplification limits the read length:
  - PCR is limited to thousands of nucleotides
  - After hundreds of synthesis cycles, clusters lose sync (sequencing out of phase)

- Not all DNA amplifies with equal efficiency (e.g. bias against CG-rich DNA)

- Third-generation sequencing uses single molecules, **without amplification**

Long-read sequencing platforms

PacBio

Oxford NANOPORE Technologies™



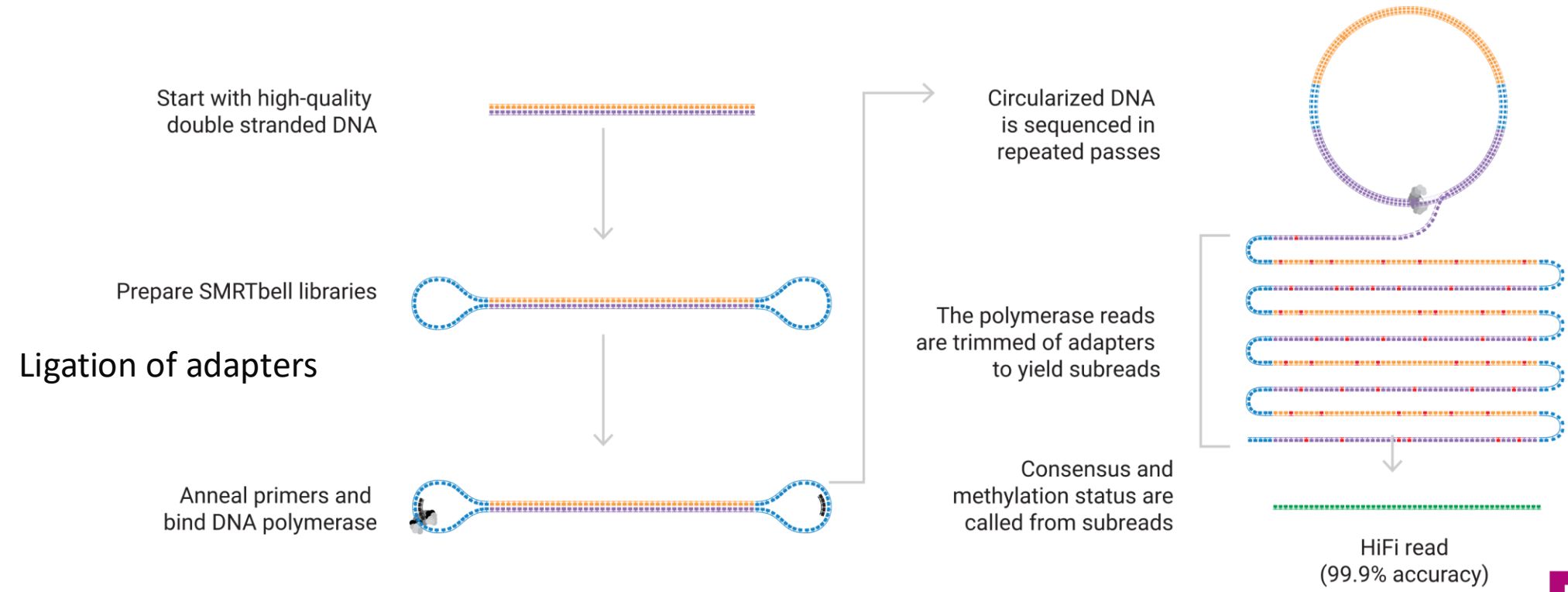
Revio

PromethION

PacBio single-molecule, real time (SMRT)

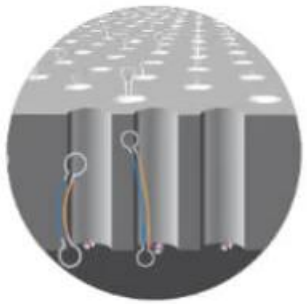
- "Principle of sequencing while synthesizing"
- Sequencing lengths up to 30kb, throughputs up to 20 Gb

Sequencing takes place in millions of tiny wells (*zero-mode waveguides)*



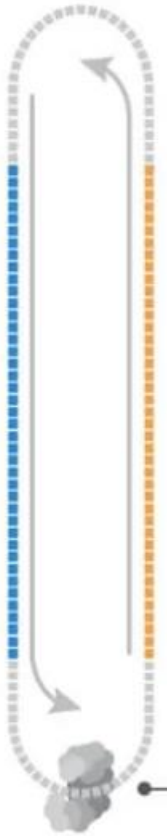
PacBio

Single Molecule Real-Time (SMRT) Sequencing Principle

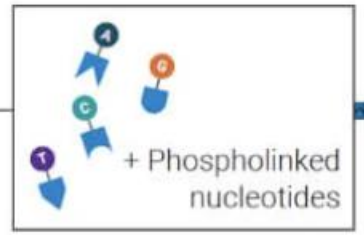


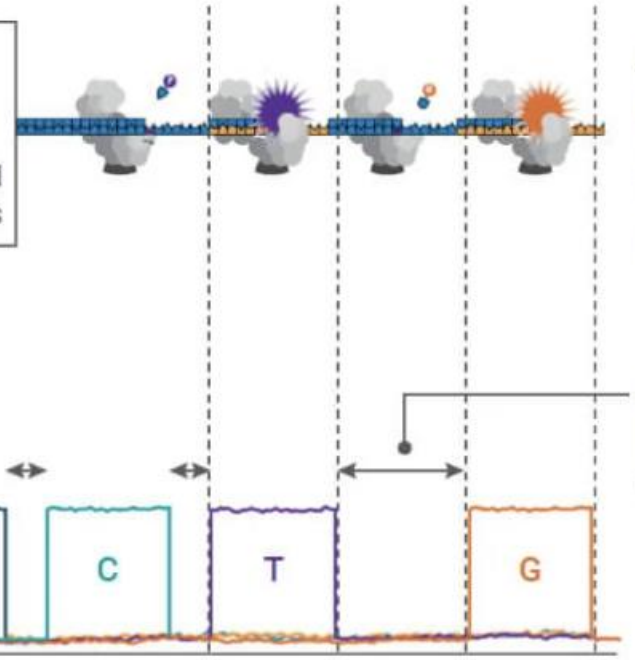
SMRT® Cells contain millions of zero-mode waveguides (ZMWs)

SMRTbell templates enable repeated sequencing of circular template with real-time detection of base incorporation



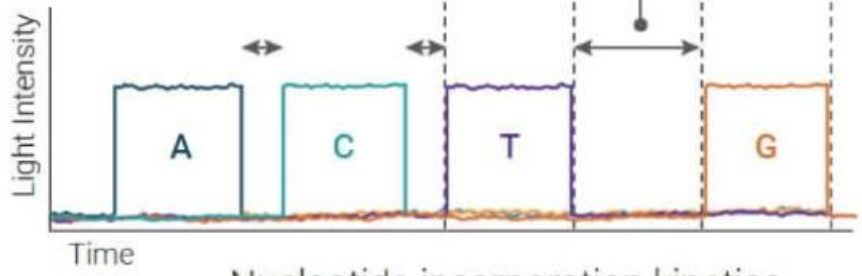
A single molecule of DNA is immobilized in each ZMW





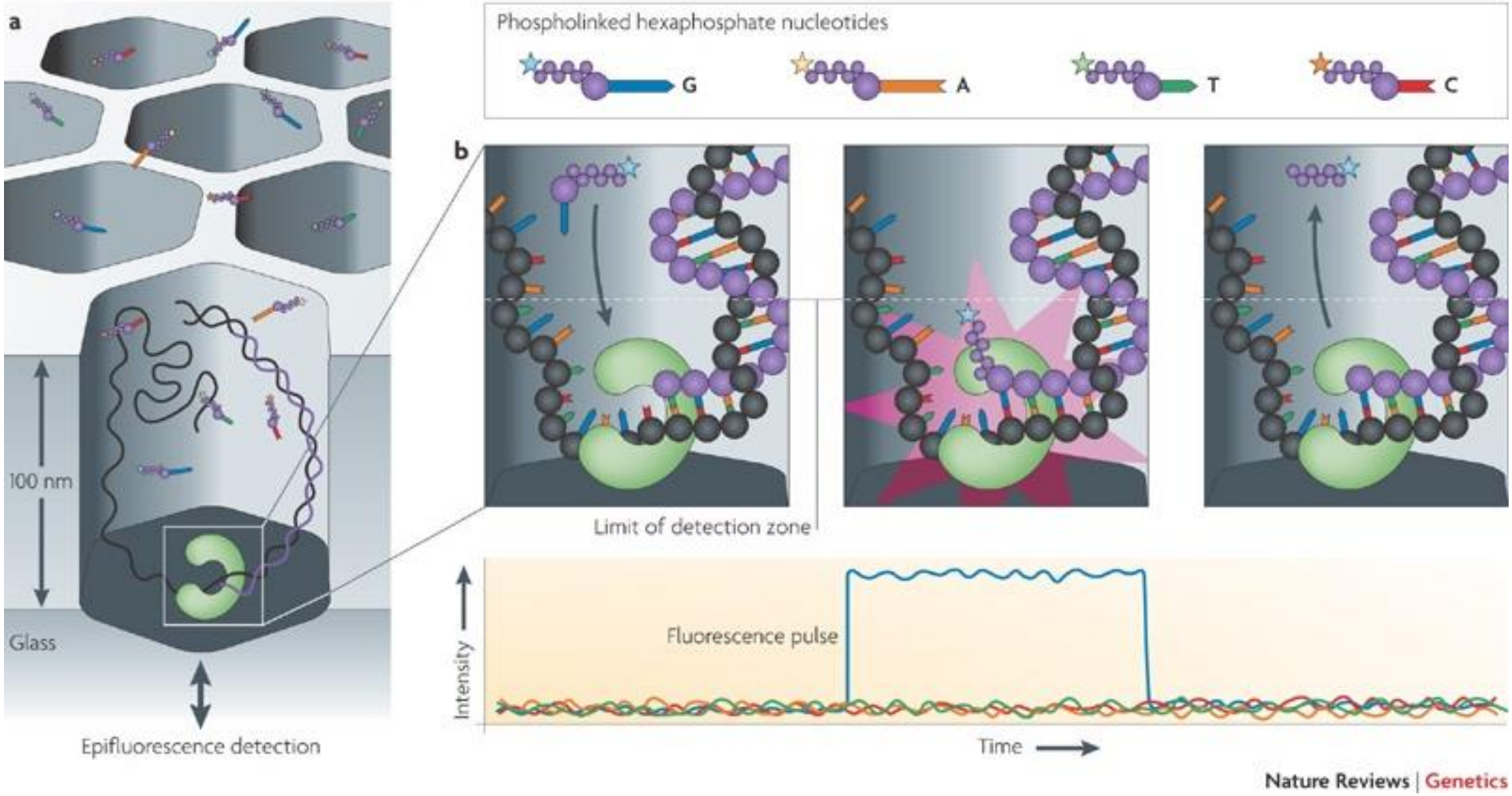
As anchored polymerases incorporate labeled bases, light is emitted

Directly detect DNA modifications during sequencing

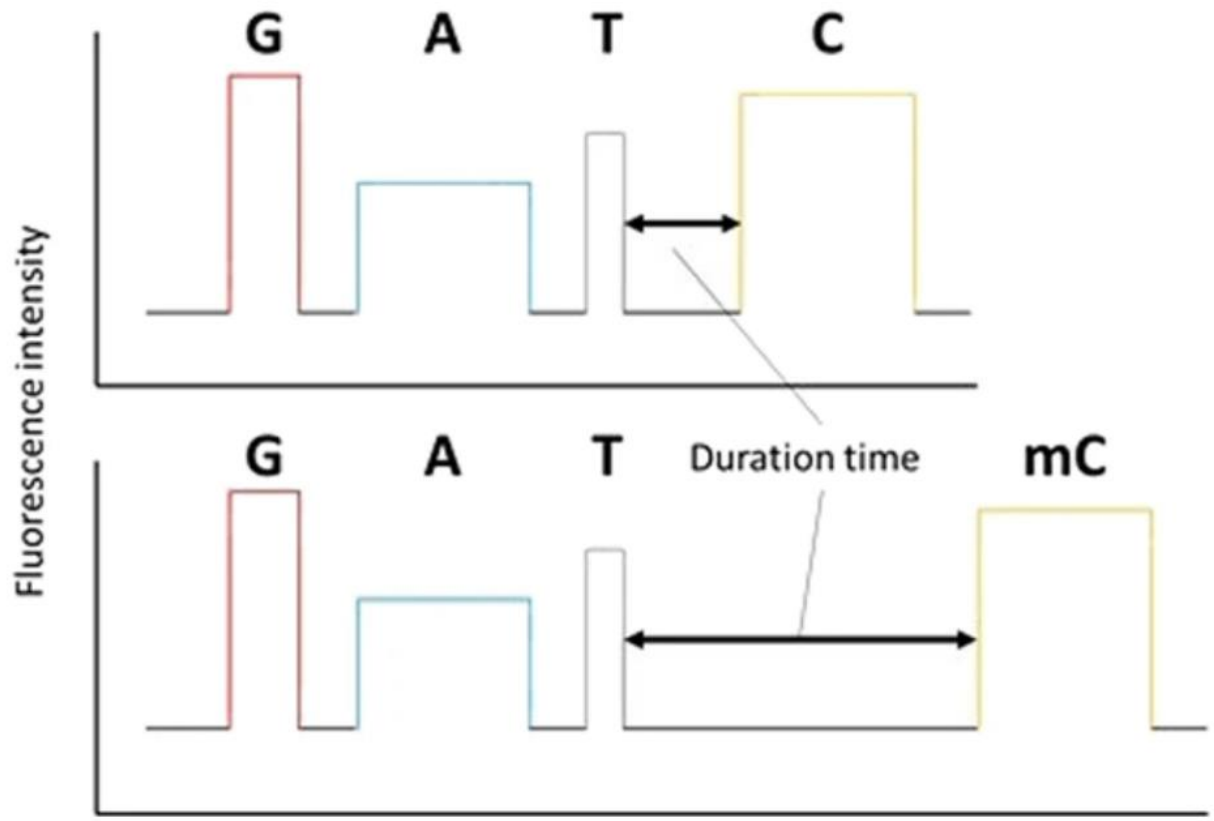


Nucleotide incorporation kinetics are measured in real time

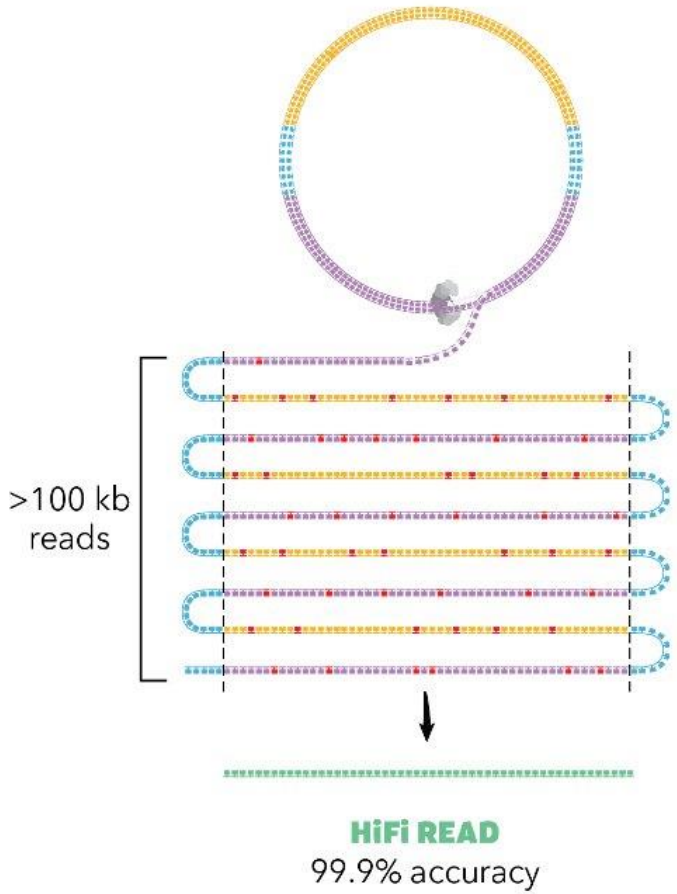
PacBio single-molecule, real time (SMRT)

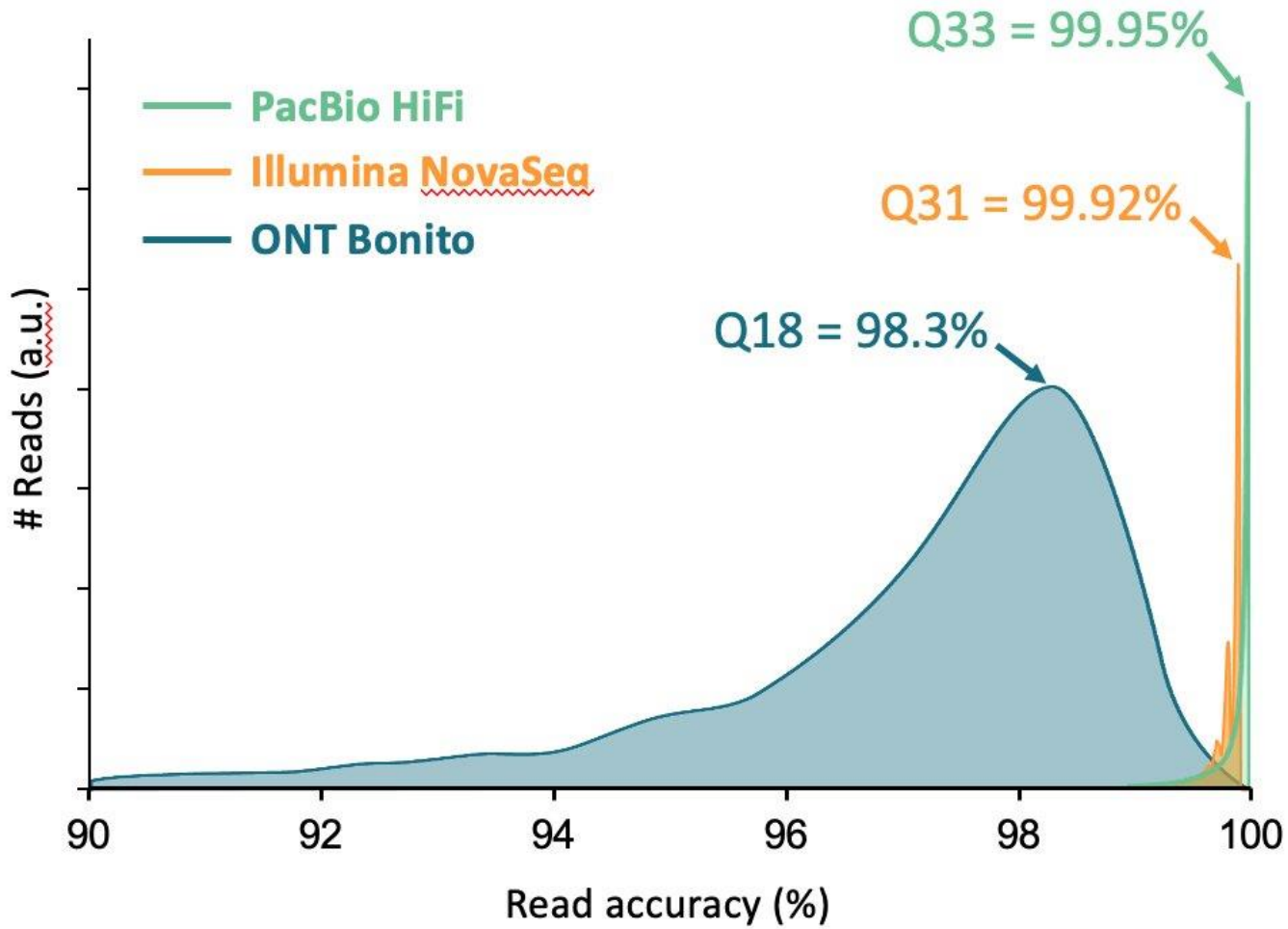


Pacbio can use polymerase kinetics to detect base modifications



PacBio HiFi Reads are Transforming Genomics





PacBio HiFi: HG003 18 kb library, Sequel II System Chemistry 2.0, precisionFDA Truth Challenge V2
Illumina: HG002 2×150 bp NovaSeq library, precisionFDA Truth Challenge V2
ONT: Bonito NCM Nanopore Tech Update Dec. 2020 and Bonito Basecalling with R9.4.1

PACBIO®

Nanopore sequencing



https://nanoporetech.com/products/sequence/minion-in-action

Sunday June 25, 1989

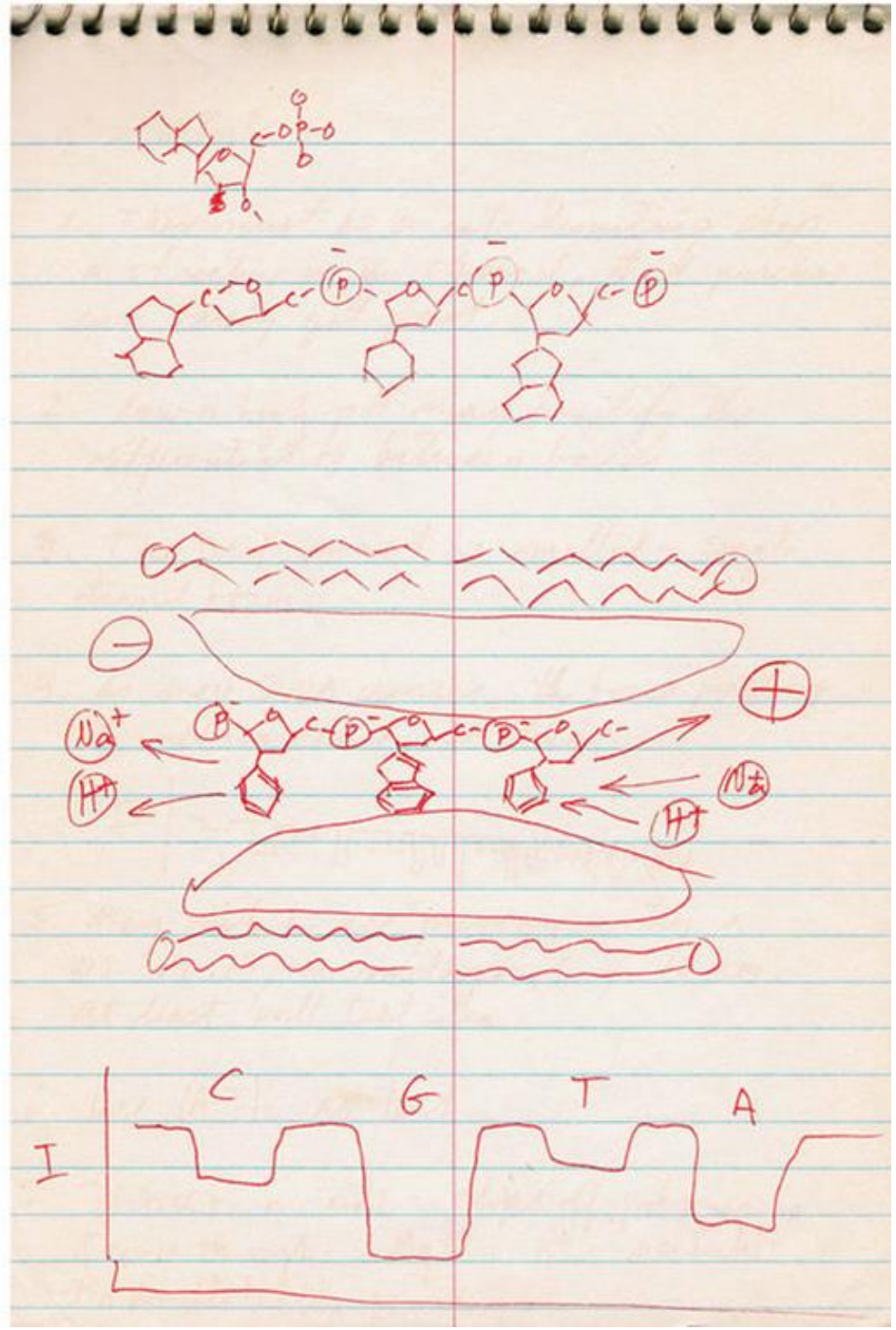
Driving back from Eugene → Belknap Lodge, had an idea on how to sequence DNA directly:

Main concept:

DNA will be driven through a small channel, either by $\Delta\Psi$ or ΔpH. The channel will be carrying a current, driven by $\Delta\Psi$. As each base passes through, a change in the current will occur. Because the bases are of different size, the current change will be proportional, thereby providing an indication of which base it is.

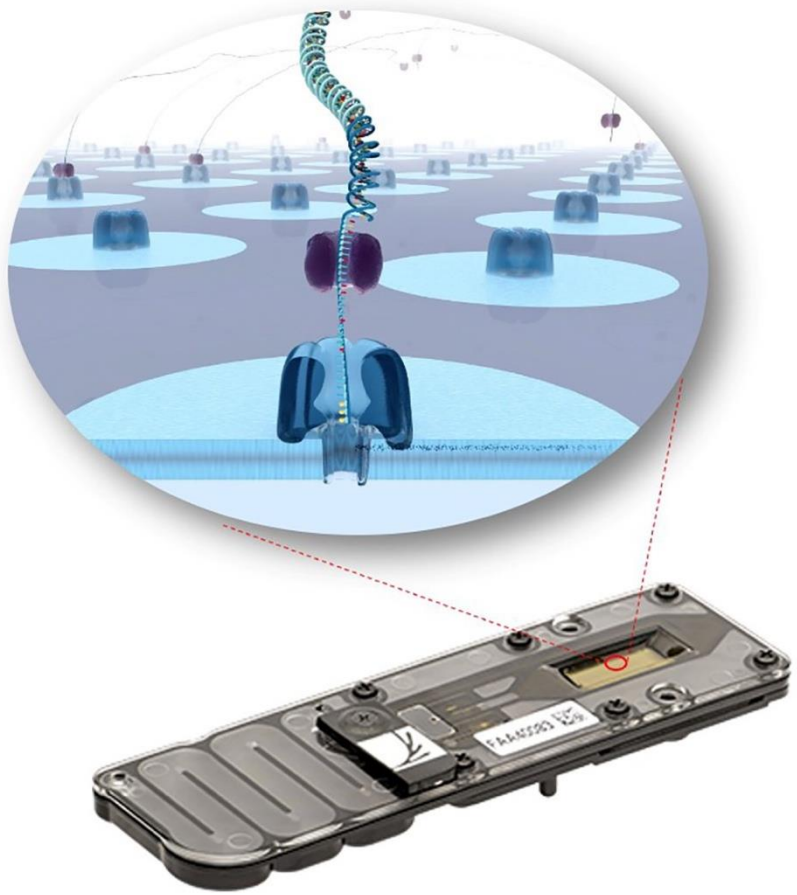
Details:

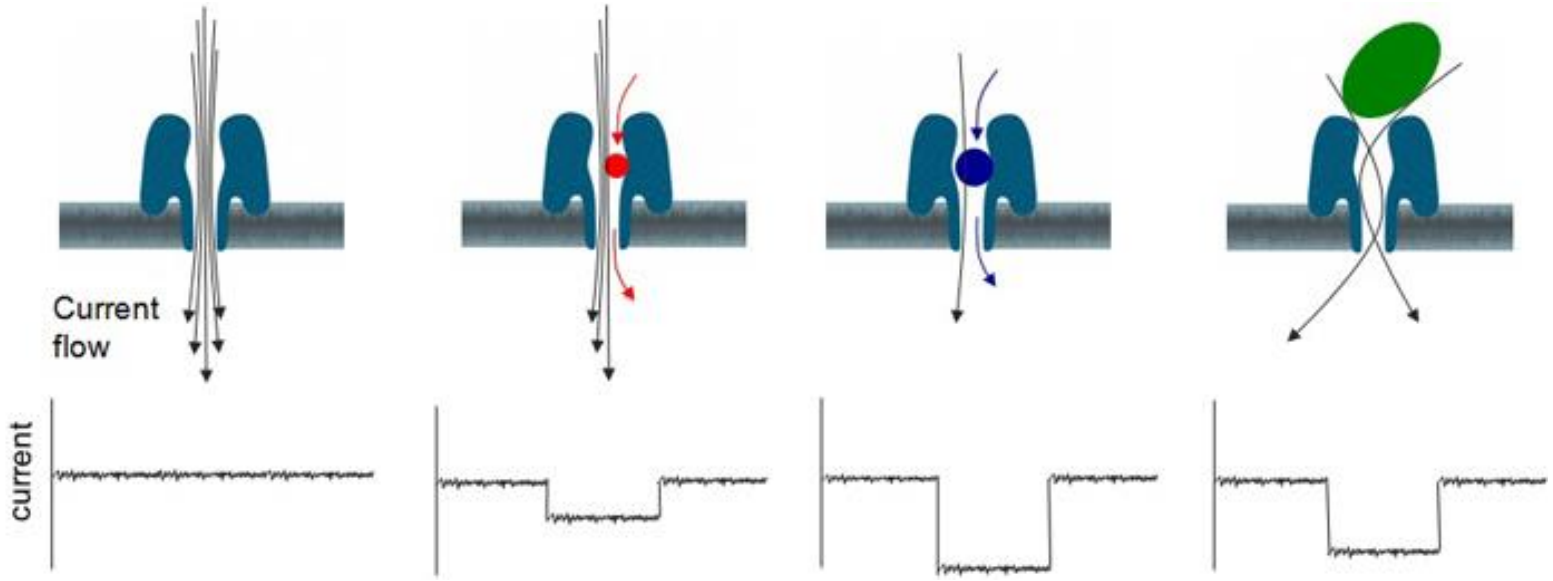
The thickness of the membrane must be very thin, perhaps a polymerized bilayer. The channel must be of the dimensions of DNA in cross section, approx. 1–2 nm. Porin? Complement? Alamethicin? The ion flux might be protonic.



Deamer's sketch in his 1989 lab notebook.

Nanopore sequencing

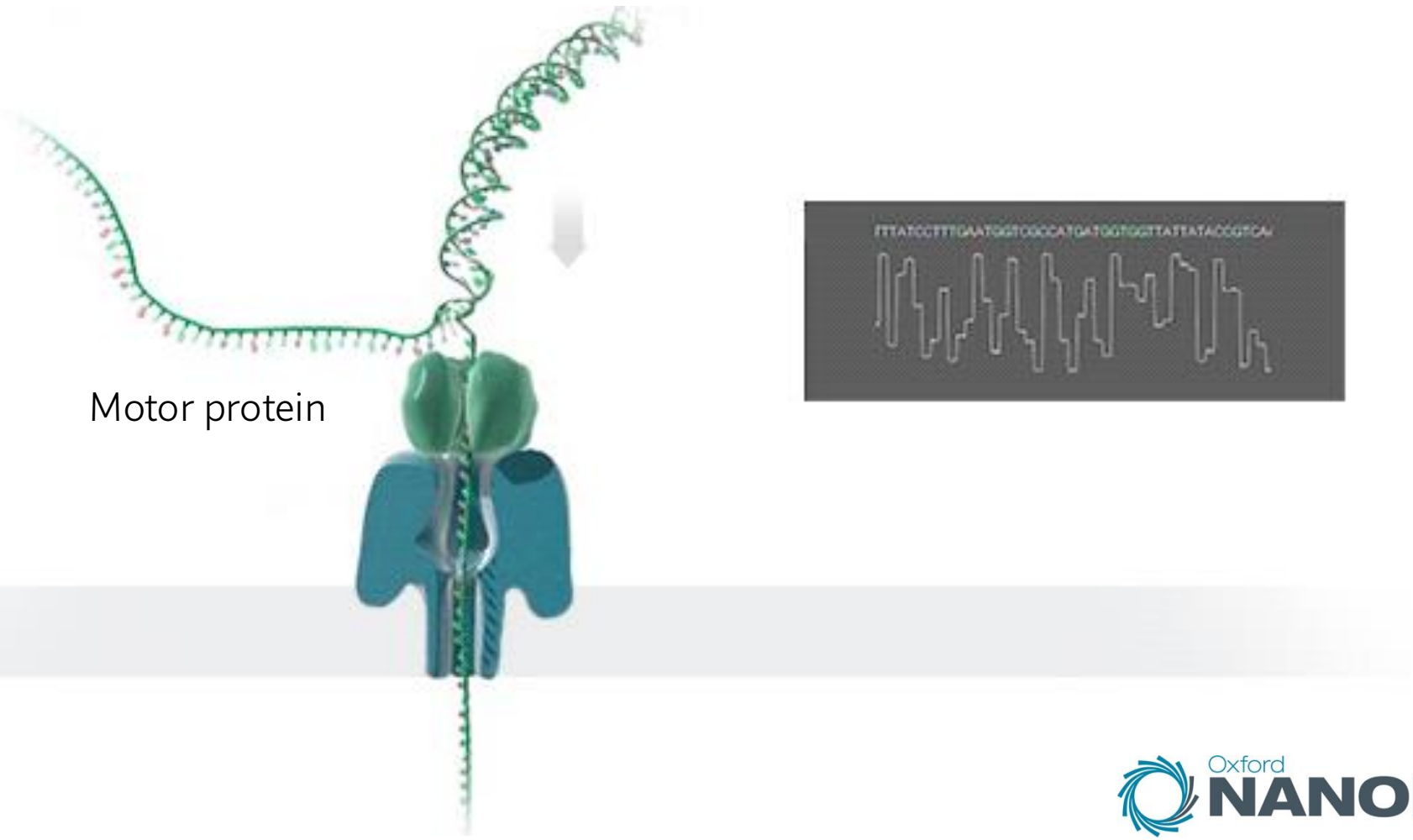




Oxford NANOPORE™ Technologies

Nanopore sequencing

Flow cell



Oxford NANOPORE™ Technologies



- MinION
- Free device, flowcells $1000

- 512 channels
- ~500 nt/second
- 20 Gbp/flowcell

Oxford NANOPORE™ Technologies



GridION Mk1

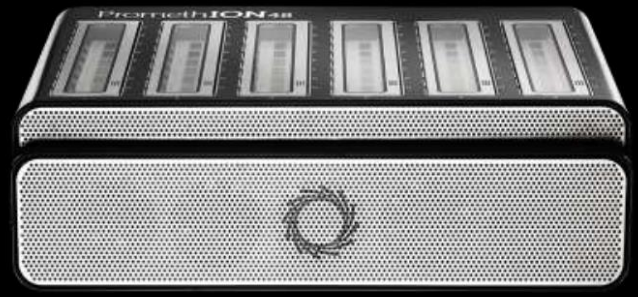


MinION Mk1C

PromethION P2



PromethION P24/P48



Flongle





PromethION P2 Solo

MinION

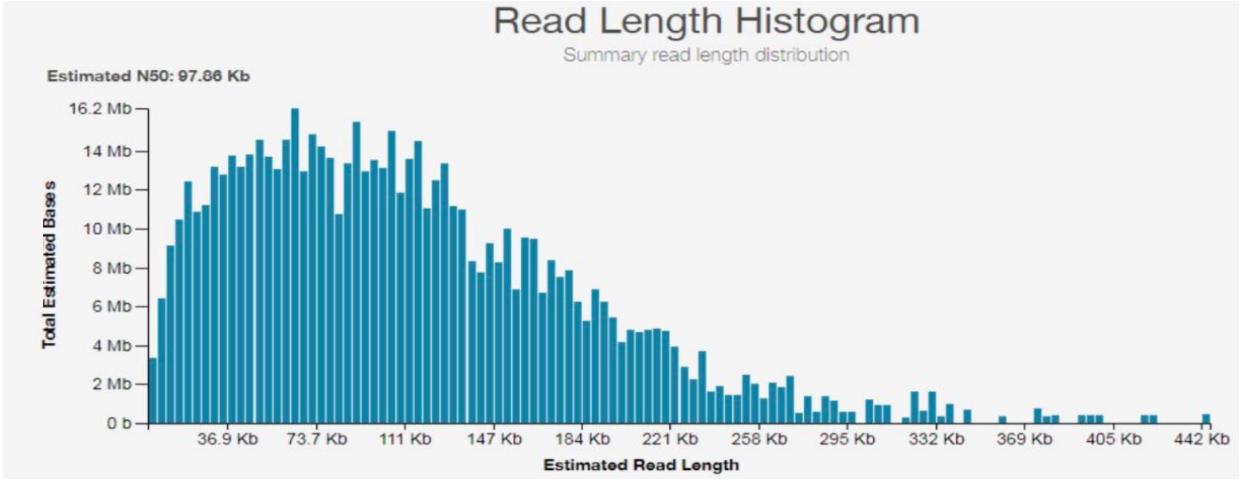




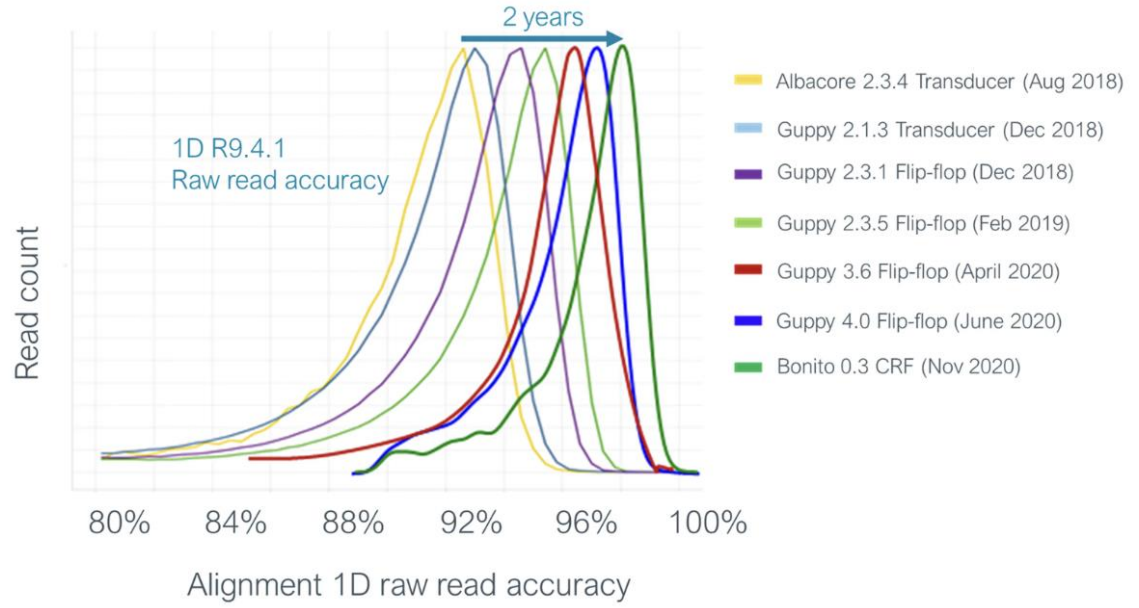
VolTRAX

Nanopore data quality

- Read lengths up to 4 Mbp (basically, depends on the DNA prep)
- No theoretical limit of length – except quality of DNA

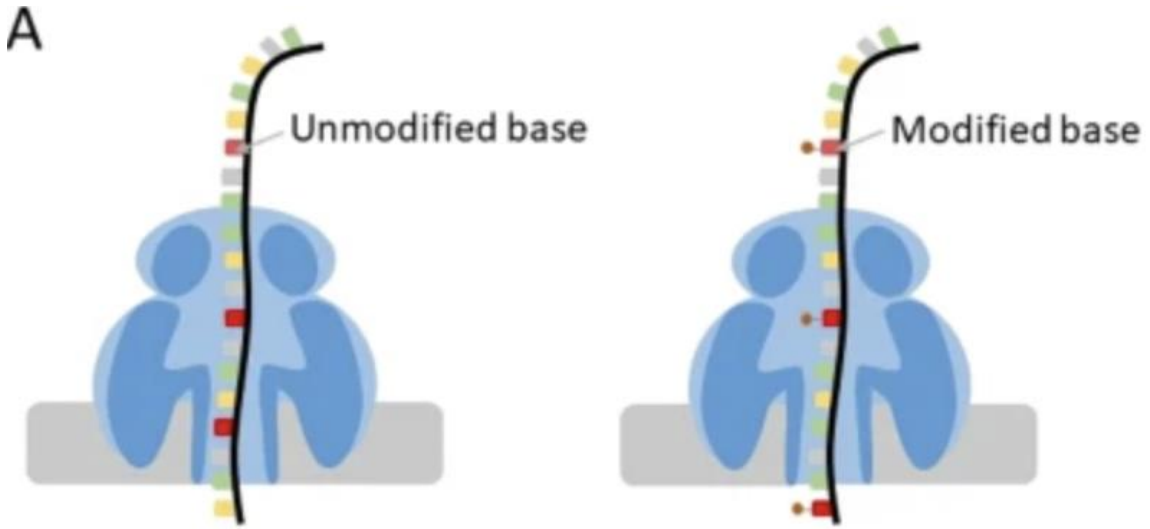


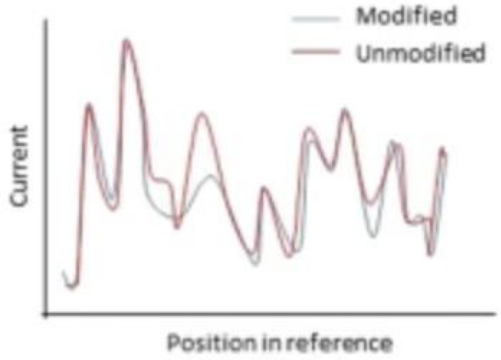
- Translation of currents to sequences (basecalling algorithms) based on neural networks
- Read quality improving with improved basecallers



Hans Jansen, John Tyson

Methylation analysis





Xu, L., Seki, M. Recent advances in the detection of base modifications using the Nanopore sequencer. *J Hum Genet* **65**, 25–33 (2020). https://doi.org/10.1038/s10038-019-0679-0

Third-generation sequencing: no amplification

😃 Read length is not limited by amplification (but by other factors)

🙂 Less CG-bias (but some other biases appear)

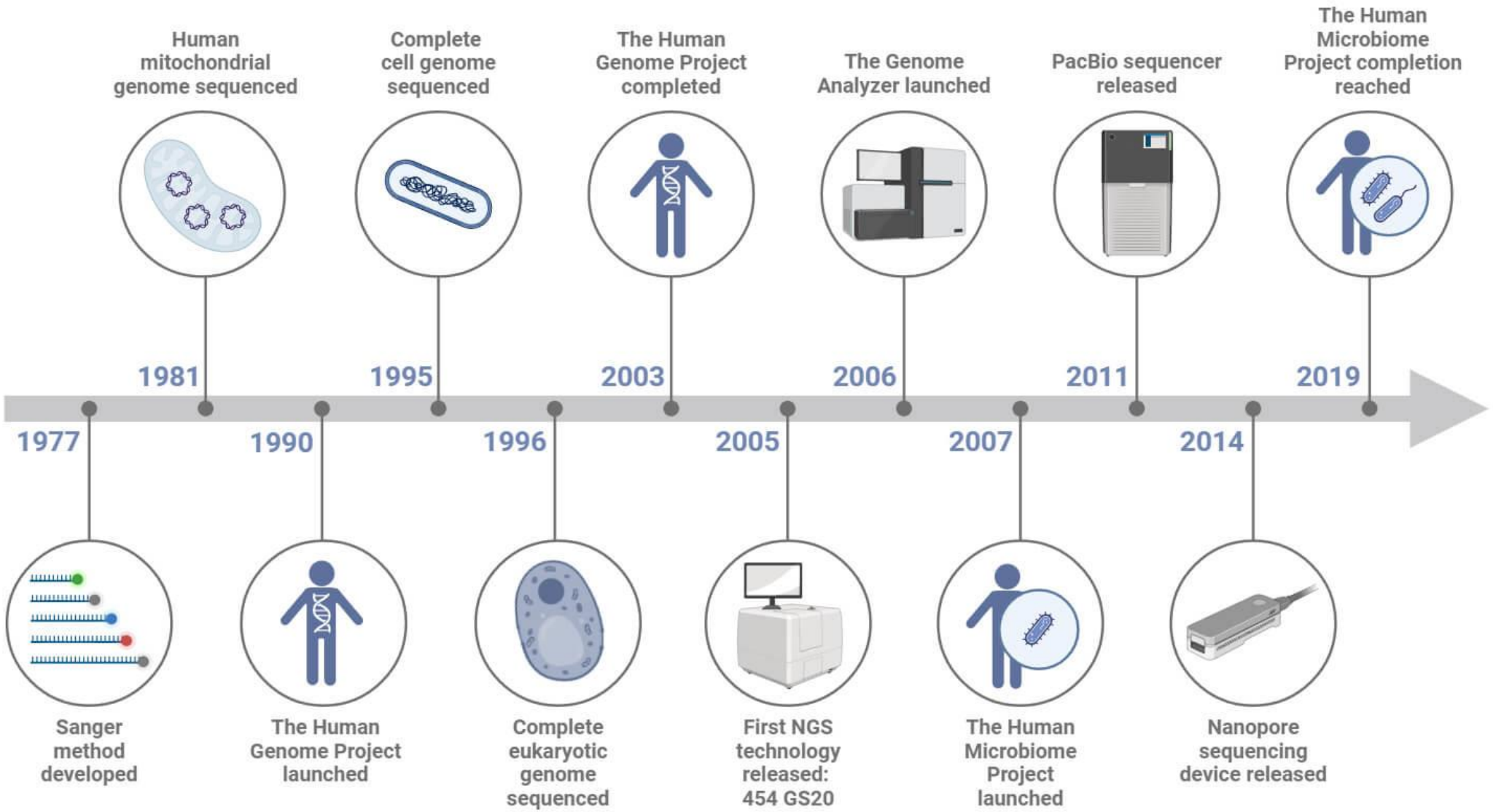
😕 Single molecules are harder to detect
- Advanced optics/sensors
- Fewer molecules simultaneously (lower read numbers)
- Higher sequencing error

🙂 High consensus accuracy: read the same molecule multiple times
- PacBio HiFi: 1 error per 1000 bp (Q30)
- Oxford Nanopore duplex (Q20+)

Summary of sequencing techologies

	Sanger	Illumina	PacBio	Oxford Nanopore
Technology	Dideoxy chain termination	Sequencing by synthesis	Single molecule, real-time (SMRT)	Nanopore sensing
Typical read length	<900	2×150	25 kbp HiFi	10 kbp – 4 Mbp
Data/flowcell	<2.88 Mbp per day	7.5 Tbp	100 Gbp	370 Gbp
Run time	A few hours	13–48 hours	<30 hours	<72 hours
Errors	<0.1%	>85% Q30	Q30 HiFi	Q20 (simplex)

The History of DNA Sequencing



Which sequencing platform to use?



Sanger sequencing is still very useful
(for instance, if you want to check a single clone)



Illumina sequencing is the NGS standard, with versatile protocols and especially useful for quantitative applications



PacBio and **Oxford Nanopore** sequencing are most useful for genome sequencing and structural studies (isoforms, structural genomic variation)